The operator-algebra arc of the GeoVac framework: a reading guide for spectral triples, state-space Gromov–Hausdorff convergence, and the Lorentzian question

J. Loutey*

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Abstract

We provide a unified reading of the fourteen papers that make up the operator-algebra arc of the GeoVac framework (Papers 29, 32, 38, 39, 40, 42, 43, 44, 45, 46, 47, 48, 49, 50). The arc constructs an explicit almost-commutative spectral triple $(\mathcal{A}_{\mathrm{GV}}, \mathcal{H}_{\mathrm{GV}}, D_{\mathrm{GV}})$ on the Fock-projected unit three-sphere with a Camporesi–Higuchi spinor Dirac operator, places it in the Marcolli–van Suijlekom gauge-network lineage, and establishes four load-bearing results. First, the underlying truncated graphs are Ramanujan *at small sizes* — a finite-size graph-RH Observation (two of four families cross the irregular Ramanujan bound at modest sizes, through real eigenvalues, Paper 29 §7), with integer-algebraic Ihara zeros (Paper 29). Second, the Connes–van Suijlekom truncations $\mathcal{T}_{n_{\max}}$ converge to the continuum spectral triple $\mathcal{T}_{S^3}$ in the van Suijlekom state-space Gromov–Hausdorff sense — for the commutative (S^3) factor on the true Camporesi–Higuchi window; convergence of the almost-commutative extension is **[OPEN]** — with the Fejér-kernel bound $d_{\mathrm{GH}}(\mathcal{T}_{n_{\max}}, \mathcal{T}_{S^3}) \leq \gamma_{n_{\max}}$, $\gamma_{n_{\max}} \sim (4/\pi)\log n_{\max}/n_{\max}$, and a measured $O(1/n_{\max})$ under a tapered amplitude (Paper 38, main theorem; $4/\pi$ in the dual-Coxeter normalisation of the SU(2) metric, $2/\pi$ on the unit S^3). Third, Paper 40 extends the convergence to compact connected semisimple Lie groups (conditionally for general G); whether the Fejér-moment constant is the same for every simple group is **[OPEN]**, since its rank-2 extractions used a normalisation that does not reproduce the SU(2) anchor. Fourth, the natural route to a Lorentzian convergence theorem on truncated $\mathrm{SU}(2) \times \mathrm{U}(1)_t$ Krein spectral triples — compressing to the Krein-positive subspace — is closed by a degeneracy theorem (Paper 45): the compression annihilates the spatial Dirac and the restricted Lipschitz seminorm vanishes identically. An earlier convergence claim through this device, together with the strong-form and bridge extensions built on it (Papers 46, 48, 49), is withdrawn; the Lorentzian analog of the convergence question Connes and van Suijlekom [6] deferred to "elsewhere" (answered on the Riemannian side for scalar compact groups by Gaudillot-Estrada and van Suijlekom, and for the spinor triple's commutative factor by Paper 38) remains open, now with a sharpened specification. Paper 47's de-compactification limit $T \to \infty$ survives at the norm-resolvent / spectral level. A master-theorem *sketch* (§9) subsumes Papers 38, 39, and 40 as special cases: for arbitrary $k \geq 1$ and compact connected simple Lie groups $G_1, \ldots, G_k$, the k-fold tensor product converges (state-space GH) with per-factor Fejér-moment bounds, leading constant open beyond SU(2) in Paper 40 (proved only for SU(2) and $\mathrm{SU}(2) \times \mathrm{SU}(2)$, on the lifted-state route with leading constant 1; the general-G $k = 1$ case is conditional on the per-group spin-window decomposition; the joint Lipschitz constant $C_3^{(k)} = 1$, by compression, belongs to the abandoned approximation-pair route). The Fejér-moment constant $4/\pi$ (Paper 38) is read — convention-dependently (dual-Coxeter normalisation) and as a reading, not a derivation — as the M_1 ($k = 0$ volume-slot) signature of the master Mellin engine, whose case-exhaustion (Paper 32 § VIII) is **[CONDITIONAL]**; the reading concerns the kernel's constant, not the GH rate; the $1/\pi^2$ of the Lorentzian HS-orthogonality (Paper 42 §11.7)

*Independent researcher. jloutey@gmail.com. This synthesis was prepared in an AI-augmented agentic workflow with Anthropic's Claude. All design decisions and editorial choices are the author's responsibility.

is *not* shown to be the same signature — it is the square of the posited normalisation $\beta = 2\pi/\kappa_g$. Further papers (48, 49, 50, 53) extend the arc; Papers 48–50 are recorded in §10–§10.3, and Paper 53 is summarized at the end of this abstract (Paper 52 is archived); metric-level claims among them descoped by Paper 45's degeneracy theorem are reported here in corrected form. **[ARCHIVED 2026-09-14]** Papers 46–49 have since been moved to `papers/archive/`: the model they rest on was withdrawn by Paper 45's annihilation theorem and the identified repair path carries its own verdict that it weakens the claim to convention. Nothing in the account below is retracted by that move, and each of the four retains content that is *not* refuted—Lemma 3.2's degeneracy diagnosis, the norm-resolvent arrow, the bridge's categorical design, and the cocycle-deficit algebra respectively. Paper 45 itself remains live and load-bearing. See `docs/retired_papers.md`. **[ARCHIVED 2026-09-28]** Paper 43 has likewise been archived: both of its headline readings were withdrawn (the finite-cutoff period closure is automatic, and its $1/\pi^2$ is the posited $\beta = 2\pi/\kappa_g$ squared), and its surviving content — the Krein carrier, the $(4, 6)$ axiom audit, the Krein-carrier period closure, the six-witness statement by construction, and the chirality-parity HS-orthogonality — now lives in Paper 42 §11, to which every Paper 43 reference below is repointed. Paper 50 re-derives the Klebanov–Pufu–Safdi sphere F-theorem values in closed Hurwitz form from the full continuum spectra (the finite truncation never enters, so this is a re-derivation of known continuum values, not a finite-cutoff result); an earlier positioning paper that read the truncation convergence as a non-holographic "correspondence" producing these values is archived (2026-09-29), its content superseded by a short not-holographic remark in Paper 50 (the framework has no bulk); and Paper 53 proves state-space convergence of the spectral truncations of the flat plane $\mathbb{R}^2$ (a positive squared-window reconstruction, rate $\gamma_1 \Lambda^{-1/2}$; the four compression/lift facts and the GH bound on the named state classes **[INTERNAL THEOREM]**, their canonicity open), leaving the disk-with-cone open: the finite-disk Markov–Cesàro reconstruction map is not positive; the plane Bochner–Riesz lift is not positive either (its kernel is negative somewhere, which is why the squared window is used); and the cone has no translation group. Paper 48 constructs Krein-pointed proper quantum metric spaces and a bridge functor to the Mondino–Sämann synthetic-Lorentzian pre-length spaces; the metric-level inheritance is descoped and the bridge survives as a conditional design. Paper 49 builds an OSLPLS (operator-system Lorentzian pre-length space) category and reads off-orbit super-additivity as the operator-algebraic dual of the twin paradox, with the deficit quantified by Datta's max-divergence chain inequality; the Λ-inheritance is descoped while the cocycle-deficit algebra survives. Paper 50 evaluates the standard zeta-regularised CFT-on-sphere free energies on S^3 and S^5 from the continuum spectra in closed Hurwitz form (re-deriving known values), exhibiting an orthogonal $\log 2$ / $\zeta(3)/\pi^2$ decomposition of the continuum (scalar, Dirac) values (an M_3-ring period and an $M_1 \times M_3$ cross-product in the master Mellin engine's period-ring stratification) and a $\log 3$ cancellation theorem on S^5. Because these values are computed from the continuum spectra, they are not evidence from the truncated triple; the framework's substrate-level scale-freedom—packing diameter d_0 cancels out, scales enter only through the Fock focal length p_0—parallels the scale-invariance of conformal field theory, but does not explain these closures. This synthesis gathers the statements, sketches the load-bearing proofs, and exhibits the structural relationships between the papers; no new theorems are introduced.

Status. The metric-level Lorentzian convergence claims of this arc are withdrawn: the K^+-compression device underlying Papers 45–46 and the bridge constructions of Papers 48–49 is degenerate (the compressed Lipschitz seminorm vanishes identically; Paper 45's annihilation theorem, §7). What stands: the unconditional Riemannian convergence theorem in van Suijlekom's state-space framework (Paper 38, translation-seminorm metrization), the operator-system substrate (Paper 44), the norm-resolvent de-compactification arrow (Paper 47), and the cocycle-deficit algebra and OSLPLS category design (Paper 49). Each affected section below carries a one-line status flag; the Status notes in the papers of record and the claims register (`docs/claims_register.md`) are authoritative for the full history.

1 Introduction

The GeoVac framework is a discretization scheme for separable quantum systems on natural geometries. When the geometry is the round $S^3 = \mathrm{SU}(2)$, the framework produces a sequence of finite truncations whose continuum limit is the Camporesi–Higuchi spectral triple of the round three-sphere [3]. The present paper synthesizes the eight documents in the project's operator-algebra group: Papers [29–36] (the former Paper 43 is now Paper 42 §11). Together they form a self-contained research thread in mathematical operator algebras with four headline results.

Headline 1: the Hopf graphs are Ramanujan at small sizes. The finite Fock-projected S^3 Coulomb graphs and the Bargmann–Segal S^5 holomorphic graphs satisfy the Kotani–Sunada Ramanujan bound at small cutoffs — a *finite-size* graph-RH: two of four families (scalar S^5, Dirac-S^3 Rule B) cross the bound at modest sizes, while the scalar S^3 graph stays below it on every connected component through $n_{\max} = 5$ (Paper 29 §7). Their *reciprocal* Ihara zeros (Hashimoto T-eigenvalues) are algebraic integers; the Ihara zeros themselves are algebraic over $\mathbb{Q}$ with integer-coefficient (but non-monic) minimal polynomials, and no π or transcendental content appears in the Ihara polynomial factorizations (Paper 29 [29], Observation [Ramanujan] and the closed-form Ihara factorizations of §5).

Headline 2: Riemannian state-space Gromov–Hausdorff convergence, with a Fejér-kernel bound of constant $4/\pi$. The Connes–van Suijlekom truncations $\mathcal{T}_{n_{\max}} = (\mathcal{O}_{n_{\max}}, \mathcal{H}_{\mathrm{GV}}, D_{\mathrm{GV}})$ of the SU(2) Camporesi–Higuchi spectral triple converge to the continuum triple $\mathcal{T}_{S^3}$ in the van Suijlekom state-space Gromov–Hausdorff sense (for the commutative factor on the true window; the almost-commutative extension is **[OPEN]**), with

$$d_{\mathrm{GH}}(\mathcal{T}_{n_{\max}}, \mathcal{T}_{S^3}) \;\le\; \gamma_{n_{\max}}, \qquad \gamma_{n_{\max}} \sim \frac{4}{\pi}\,\frac{\log n_{\max}}{n_{\max}} \quad (n_{\max} \to \infty). \tag{1}$$

Here $\gamma_{n_{\max}}$ is the first moment of the central Fejér kernel; a tapered amplitude gives a measured $O(1/n_{\max})$, so $4/\pi$ is the kernel's constant, not the convergence rate (Paper 38, main theorem). That constant is the sphere-volume quotient $2\mathrm{Vol}(S^2)/\mathrm{Vol}(S^3)$ at unit radius (numerically $\mathrm{Vol}(S^2)/\pi^2$) — read (a reading, not a derivation; Paper 38) as the $k = 0$ volume factor of the master Mellin engine, not the Hopf fibration's base-to-total ratio (Paper 38, volume-ratio note; the fibration is $S^1 \to S^3 \to S^2$) — and is discussed in detail in §4–§8 below (Paper 38 [31], Theorems 3.4 and 4.1).

Headline 3: the Fejér-moment constant beyond SU(2) is open. For a simple compact Lie group G with bi-invariant metric, Paper 40 proposes a Plancherel-weight × Vandermonde-Jacobian cancellation mechanism (§3.2 [33]; a proof sketch) under which the leading constant would be the rank-1 one; whether the constant is group-independent is **[OPEN]**, because the rank-2 extractions used a normalisation that does not reproduce the SU(2) anchor. The Lipschitz comparison constant $C_3 = 1$ (which enters only the withdrawn approximation-pair route) holds at every rank by compression (Paper 40's PRV-summand analysis after Kumar (1988), applied to the dominant image of the tensor product $V_\lambda \otimes V_{\lambda'}^*$, concerns representation-theoretic inequalities, proved on special summands and panel-verified on the rest, not this bound). *Tier:* this universality is derived at rank 1 (SU(2)) and, for general G, is a proof-sketch mechanism (the Plancherel×Vandermonde cancellation) numerically supported, with a fit-sensitive extraction, at rank 2 ($\mathrm{SU}(3), \mathrm{Sp}(2), G_2$) with one rank-3 datapoint; a fully rank-uniform analytical proof is a named gap (Paper 40 §L2), and the per-group constant extractions are fit-sensitive.

Headline 4: the Lorentzian propinquity route is closed by a degeneracy theorem. On truncated $\mathrm{SU}(2)\times\mathrm{U}(1)_T$ Krein spectral triples $\mathcal{T}^L_{n_{\max},N_t,T}$ built via the Bizi–Brouder–Besnard

[2] signature $(3,1)$ data at $(m,n) = (4,6)$, the natural K^+-restricted weak-form Lorentzian propinquity target bound would be

$$\Lambda^L(\mathcal{T}^L_{n_{\max},N_t,T}, \mathcal{T}^L_{\mathcal{M}}) \;\leq\; C_3^{\text{joint}} \cdot \gamma^{\text{joint}}_{n_{\max},N_t,T} \;\longrightarrow\; 0, \tag{2}$$

with $\gamma^{\text{joint}}_{n_{\max},N_t,T} = O(\log n_{\max}/n_{\max} + T \log N_t/N_t)$, where $\Lambda^L(\cdot,\cdot) := \Lambda(\cdot|_{\mathcal{K}^+}, \cdot|_{\mathcal{K}^+})$ is the propinquity of the standard metric spectral triple obtained from the Krein-positive subspace $\mathcal{K}^+ = \{|\psi\rangle : J_L|\psi\rangle = +|\psi\rangle\}$. The target fails structurally: the quantity Λ^L is degenerate — the K^+ compression annihilates the spatial Dirac and the restricted Lipschitz seminorm vanishes identically (Paper 45's annihilation theorem [36]) — and the earlier convergence claim through this device (formerly Theorem 5.1 of Paper 45) is withdrawn. The Lorentzian analog of the convergence question Connes and van Suijlekom [6] deferred to "elsewhere" (answered on the Riemannian side for scalar compact groups by Gaudillot-Estrada and van Suijlekom, and for the spinor triple's commutative factor by Paper 38) remains open, now with a sharpened specification.

Position against the literature. The arc sits in the lineage of Connes–van Suijlekom 2021 [6], which formalised spectral truncations as operator systems with finite-cutoff Connes distance; of Marcolli and van Suijlekom 2014 [16], which introduced gauge networks as a discrete-graph realisation of finite spectral triples carrying connection on edges; and of the subsequent Latrémolière–Farsi propinquity programme and the Hekkelman–Leimbach–Toyota work on Gromov–Hausdorff convergence of spectral truncations (Hekkelman–McDonald's noncommutative integral on truncations [12] is a related, non-convergence result) [7, 8, 11, 13, 15, 25]. All these earlier convergence results are Riemannian or discrete-group (Toyota's are for discrete groups of polynomial growth), and almost all of the Riemannian ones are on flat structures (S^1, $\mathbb{T}^d$, S^2 Berezin–Toeplitz). The Paper 38 SU(2) result and the Paper 40 generalisation (for general G conditional on the per-group spin-window decomposition) are non-abelian non-flat instances at quantitative rate (Paper 38 unconditional for the commutative factor, via the translation-seminorm metrization; scalar-sector convergence for compact groups is due to Gaudillot-Estrada and van Suijlekom, arXiv:2310.14733); on the Lorentzian side the Paper 45 contribution is a degeneracy theorem, not a convergence instance. Mondino and Sämann's synthetic Lorentzian Gromov–Hausdorff program [17, 18] is categorically distinct in object class (pre-length spaces with causal diamonds, not operator-algebraic spectral triples) and is not in competition with the present arc.

What this synthesis provides. A working mathematician reading the corpus cold may find the nine-paper sequence difficult to navigate. This document collects the constructions, restates the load-bearing theorems with cross-references back to the original papers, and exhibits the relationships between them. No new theorems are introduced. The reader is assumed familiar with spectral triples [4, 5], Lipschitz seminorms and Rieffel quantum metric spaces, and the Latrémolière propinquity [13, 14]. Familiarity with [6] is helpful but not strictly required. No knowledge of the GeoVac physics-side corpus (quantum chemistry, QED, precision spectroscopy) is assumed; this arc is self-contained from the operator-algebra side.

2 The GeoVac spectral triple

2.1 The Camporesi–Higuchi data on S^3

We work throughout on the unit three-sphere S^3, identified with SU(2) under its bi-invariant round metric. Following Camporesi and Higuchi [3], the spinor bundle is parallelisable, of rank 2, and the Dirac operator D_{CH} has spectrum

$$\text{spec}(D_{\text{CH}}) = \{\pm(n+\tfrac{3}{2}) : n \in \mathbb{N}_0\}, \qquad g_n^{\text{Dirac}} = 2(n+1)(n+2) \tag{3}$$

on each chirality of the half-integer Hurwitz-shifted shell. We write $\mathcal{H}_{S^3} = L^2(S^3, \Sigma)$ for the chirality- doubled spinor Hilbert space carrying the Camporesi–Higuchi data.

2.2 The GeoVac algebra, Hilbert space, and Dirac operator

The GeoVac construction (Paper 32 §3 [30]) takes the following data:

- $\mathcal{A}_{\mathrm{GV}} \subset C^\infty(S^3)$ is the involutive algebra of complex-valued smooth functions on S^3, with the standard pointwise product and complex conjugation;
- $\mathcal{H}_{\mathrm{GV}} = \mathcal{H}_{S^3}$, with $\mathcal{A}_{\mathrm{GV}}$ acting by pointwise multiplication;
- $D_{\mathrm{GV}} = D_{\mathrm{CH}}$, the Camporesi–Higuchi Dirac operator;
- J_{GV} is the charge-conjugation real structure inherited from the spin structure of S^3, satisfying $J_{\mathrm{GV}}^2 = -I$ (Euclidean KO-dimension 3), $J_{\mathrm{GV}} D_{\mathrm{GV}} = +D_{\mathrm{GV}} J_{\mathrm{GV}}$.

The triple $(\mathcal{A}_{\mathrm{GV}}, \mathcal{H}_{\mathrm{GV}}, D_{\mathrm{GV}}; J_{\mathrm{GV}})$ is real, odd (no $\mathbb{Z}_2$-grading; KO-dimension 3), and has compact resolvent $|D_{\mathrm{GV}}|^{-1}$ with eigenvalue multiplicities given by (3). The Connes axiom audit is recorded in Paper 32 §4 [30], where each of the seven axioms (order zero, bounded commutators, compact resolvent, order one, real structure, $\mathbb{Z}_2$-grading by convention, and finite summability) is checked.

2.3 The Marcolli–van Suijlekom lineage

The role of GeoVac in the ecosystem of finite-and-graph spectral triples is explained in Paper 32 §7. The Marcolli–van Suijlekom construction [16] attaches a finite spectral triple to each vertex of a graph and a connection (gauge group element) to each edge; the spectral action of the resulting object recovers Wilson lattice gauge theory. The continuum-limit issue addressed by Perez-Sanchez [20] clarifies that the limit is Yang–Mills without Higgs by default; a Higgs sector appears only via off-diagonal data on the finite factor.

The GeoVac Fock-projected S^3 graph fits this template naturally. Its vertices are the Fock index (n, ℓ, m) with $n = 2j + 1$, ℓ the orbital, and m the magnetic quantum number, providing a Peter–Weyl-indexed basis on $S^3 = \mathrm{SU}(2)$. Each Fock shell carries a finite spectral triple (roughly, a finite-dimensional subrepresentation of the Camporesi–Higuchi triple), and the edge structure encodes the SO(4) selection rules of the underlying graph. The graph admits a SU(2) Wilson construction, natural because $S^3 = \mathrm{SU}(2)$, and its maximal-torus restriction, compact $U(1)$ (both exactly solvable on the scalar graph, Paper 30), and an SU(3) Wilson construction on the Bargmann–Segal S^5 [28], all of which carry the standard kinetic coefficient $1/(4N_c)$.

The Connes axiom audit (Paper 32 §4 [30]) checks explicitly that $J^2 = -I$ and $JD = +DJ$ hold on the truthful Camporesi–Higuchi data at every finite cutoff $n_{\max} \in \{1, 2, 3\}$ tested. These two relations, together with the parallelisability of $S^3 = \mathrm{SU}(2)$, distinguish the GeoVac triple from the Connes–van Suijlekom circulant comparator, which is commutative and gives propagation number 1.

2.4 Truncations as operator systems

For each $n_{\max} \in \mathbb{N}$, the orthogonal projection $P_{n_{\max}} : \mathcal{H}_{\mathrm{GV}} \to \mathcal{H}_{\mathrm{GV}}$ onto the span of Peter–Weyl modes with $n \le n_{\max}$ defines a truncated operator system

$$\mathcal{O}_{n_{\max}} \;:=\; P_{n_{\max}}\, \mathcal{A}_{\mathrm{GV}}\, P_{n_{\max}} \;\subset\; \mathcal{B}(P_{n_{\max}} \mathcal{H}_{\mathrm{GV}}), \tag{4}$$

which is self-adjoint, contains the identity, but is in general *not* multiplicatively closed. The defect is precisely the content of the Connes–van Suijlekom operator-system formalism [6]: in the natural envelope, $\mathcal{O}_{n_{\max}}$ has propagation number $\mathrm{prop}(\mathcal{O}_{n_{\max}}) = 2$ at every tested $n_{\max} \in$

$\{2,3,4\}$, matching the Toeplitz S^1 result [6, Proposition 4.2] verbatim. A concrete witness pair at $n_{\max}=2$ is $M^2_{2,1,0}\notin\mathcal{O}_2$ with 14.9% Frobenius residual against $\mathcal{O}_2$, deficit concentrated on the top-shell diagonal in the natural Avery basis (Paper 32 §3.4 [30]). The truncated triple is then

$$\mathcal{T}_{n_{\max}}=(\mathcal{O}_{n_{\max}},P_{n_{\max}}\mathcal{H}_{\mathrm{GV}},P_{n_{\max}}D_{\mathrm{GV}}P_{n_{\max}}),$$

with continuum limit $\mathcal{T}_{S^3}=(\mathcal{A}_{\mathrm{GV}},\mathcal{H}_{\mathrm{GV}},D_{\mathrm{GV}})$.

3 The Hopf graphs as Ramanujan substrate

Before turning to convergence, we record the most basic spectral-graph property of the underlying finite truncations.

Ihara zeta of the Fock graph. For each $n_{\max}$, let $G_{n_{\max}}$ denote the directed Fock-projected S^3 Coulomb graph: vertices are triples (n,ℓ,m) with $n\le n_{\max}$, $\ell<n$, $|m|\le\ell$; edges encode SO(4) selection rules. The Ihara zeta function $\zeta_{G_{n_{\max}}}(s)$ is computed in Paper 29 §4 [29] via the Ihara–Bass determinantal formula and via the $2E\times 2E$ Hashimoto non-backtracking edge operator.

Remark 3.1 (Observation — Paper 29, Ramanujan bound + §7 bound-crossing)**.** At small cutoffs the Fock-projected S^3 graph $G_{n_{\max}}$ satisfies the Kotani–Sunada Ramanujan bound strictly: the largest non-trivial Hashimoto eigenvalue sits strictly inside the spectral radius $\sqrt{q_{\max}}$ associated with the maximum (Perron) vertex degree. This is a *finite-size* statement, not an asymptotic theorem: Paper 29 §7 shows two of four families (scalar S^5, Dirac-S^3 Rule B) *cross* the bound at modest sizes; the S^3 graph stays below it on every connected component through $n_{\max}=5$, and no asymptotic statement is proved for it. The reciprocal Ihara zeros (Hashimoto T-eigenvalues) are algebraic integers, while the Ihara zeros themselves are algebraic over $\mathbb{Q}$; the corresponding Ihara polynomials factor over $\mathbb{Z}[s]$ into integer-coefficient pieces, and several factorizations admit explicit closed forms (e.g. $(4s^2+1)^2$, $(9s^2+1)^4$, the $12+22$ dichotomy for S^5 at $N_{\max}=3$).

Structural consequences. The Observation 3.1 secures two facts about the truncated graphs that are used implicitly elsewhere in the arc. First, no π or transcendental quantity appears anywhere in the Ihara polynomials, in any of the tested factorizations (Paper 29 §5.4 [29], Corollary on integer-algebraicity). This is the graph-spectral analog of the π-source case-exhaustion theorem (Paper 32 §8, **[CONDITIONAL]**, first fifteen Paper 34 projections): both say that, where π appears, it enters through specific named mechanisms ($k=0$ volume slot; spectral action; vertex-parity Hurwitz character), never in the bare combinatorial data. Second, the Hashimoto-spectrum Poisson statistics (Paper 29 §6.3) are categorically distinct from the GUE-like statistics seen on the spectral-zeta side of the framework, making clear that the Weil dictionary between Selberg and Riemann breaks at GeoVac by design.

At the small cutoffs where it holds, the Ramanujan property of the truncated graphs lets one take naive sums and traces over $G_{n_{\max}}$ without pathological spectral gaps, and the Ihara polynomials are entirely in $\mathbb{Z}[s]$. It does not persist as $n_{\max}\to\infty$ (above), and Paper 38's convergence theorem does not use it.

4 Riemannian Gromov–Hausdorff convergence: the central theorem

4.1 Statement (Paper 38)

The load-bearing theorem of the arc is Paper 38 [31], Theorem 3.4.

Theorem 4.1 (Paper 38, GH convergence on $S^3 = \mathrm{SU}(2)$). *The Connes–van Suijlekom truncations $\mathcal{T}_{n_{\max}}$ of the round* SU(2) *Camporesi–Higuchi spectral triple converge to $\mathcal{T}_{S^3}$ in the van Suijlekom state-space Gromov–Hausdorff sense, for the commutative factor on the true Camporesi–Higuchi window (translation-seminorm metrization; the almost-commutative extension is **[OPEN]**):*

$$d_{\mathrm{GH}}(\mathcal{T}_{n_{\max}}, \mathcal{T}_{S^3}) \;\le\; \gamma_{n_{\max}} \;\to\; 0 \quad (n_{\max} \to \infty), \tag{5}$$

with constant 1 *from the lifted-state route, where γ_n is the first moment of the central Fejér kernel* (7)*, with **[INTERNAL THEOREM + MEASURED]** asymptotic constant $4/\pi$:*

$$n\,\gamma_n \;\sim\; (4/\pi)\,\log n \qquad (n \to \infty). \tag{6}$$

***[MEASURED]** A uniform finite-cutoff bound $\gamma_n \le 6\,(\log n)/n$ holds on the checked sample of $n \ge 2$ (not proved for all n). **[MEASURED]** With a Cesàro-tapered amplitude in place of the Fejér one, $d_{\mathrm{GH}} = O(1/n_{\max})$ (proof and optimal constant **[OPEN]**); $4/\pi$ is the Fejér kernel's constant, not the rate.*

The proof is the *lifted-state* construction, which uses the L2 central Fejér kernel and the L4(c) good-kernel estimate on the true window; the lemmas L1′ (the chirality-doubled model substrate) and L3 are sketched below as supporting infrastructure. A fifth lemma (L5, the Latrémolière propinquity assembly) was part of the original architecture and was **withdrawn on 2026-09-03** when its height bound was refuted; it is retained in Paper 38 as a record and is *not* part of the proof of Theorem 4.1. See §4.6 below for what was refuted and what carries the theorem in its place.

4.2 Lemma L1′: chirality-doubled truncated operator system

The original Connes–van Suijlekom truncation is on the scalar Fock graph; the spinor lift to the full Camporesi–Higuchi bundle (with both chiralities) is the subject of an internal sprint (R3.5).[1] L1′ concerns the doubled computational model, not the true window: on the doubled Fock spin-orbital model the Dirac-commutator seminorm has a kernel larger than the scalars, while on the true Camporesi–Higuchi window it is a norm at $n_{\max} \le 3$ (Paper 38, Rem. dirac_degeneracy). The chirality-doubled operator system has dimension $\dim \mathcal{O}_{n_{\max}} = 1, 14, 55, 140, \ldots$ at $n_{\max} = 1, 2, 3, 4, \ldots$ and every cross-shell pair has finite Connes distance. See Paper 38 §4.1 [31] for the explicit construction.

4.3 Lemma L2: central spectral Fejér kernel on SU(2)

The proof of Theorem 4.1 requires a positive, state-preserving, central good kernel on SU(2) whose mass-concentration moment vanishes at a known rate: it is this kernel that supplies both the function-level good-kernel estimate and the operator-side conjugation average Φ in the unconditional argument (§4.6). (The withdrawn L5 tunneling pair also used it; the kernel is load-bearing either way, which is why L5's withdrawal leaves it untouched.) Paper 38 §4.2 constructs the central spectral Fejér kernel

$$K_{n_{\max}}(g) \;=\; \frac{1}{Z_{n_{\max}}} \left| \sum_{j \le j_{\max}} \sqrt{2j+1}\,\chi_j(g) \right|^2, \qquad j_{\max} = (n_{\max}-1)/2, \quad Z_{n_{\max}} = \frac{n_{\max}(n_{\max}+1)}{2}. \tag{7}$$

The kernel's mass distribution on the Wen–Avery basis is $(2j+1)/Z_{n_{\max}}$ for $j \le j_{\max}$; it is *not* the multiplier symbol $\sigma_{n_{\max}}$ (Paper 38). The associated central smoothing map is unital

[1] Parenthetical sprint/track designations are anchors into the project's internal chronicle (`CHANGELOG.md` in the repository); "WH*n*" designations are entries of the internal working-hypothesis register (`CLAUDE.md` §1.7).

completely positive, so its cb-norm is

$$\|S_{K_{n_{\max}}}\|_{\mathrm{cb}} \;=\; 1 \tag{8}$$

(the value $2/(n_{\max}+1)$ printed here before 2026-09-30 is the mass-distribution maximum, not a cb-norm): convolution by a positive unit-mass kernel is unital completely positive, and a UCP map has cb-norm 1 (Paulsen [19], Prop. 3.6).

The Fejér-moment constant $4/\pi$. The mass-concentration moment $\gamma_{n_{\max}}$ of the kernel admits an exact closed-form sum rule whose asymptotics are fixed by an Euler–Maclaurin analysis of two logarithmic sub-coefficients (net $2-1=1$); the corpus names this the *Stein–Weiss sharpening*, after the real-line approximate-identity estimates of [24] (Paper 38, Appendix A):

$$\gamma_{n_{\max}} \;=\; \pi - \frac{4\,T_{n_{\max}}}{\pi\,Z_{n_{\max}}}, \quad \text{with } \frac{n\,\gamma_n}{\log n} \to \frac{4}{\pi} \text{ as } n\to\infty. \tag{9}$$

The constant is the sphere-volume quotient $\mathrm{Vol}(S^2)/\mathrm{Vol}(S^3) = 2/\pi$ at unit radius, twice that in the dual-Coxeter normalisation. Both spheres have radius 1, so this is *not* the Hopf fibration's base-to-total ratio (that base is $S^2(1/2)$ and the ratio is $1/(2\pi)$; Paper 38, volume-ratio note). Concretely $\mathrm{Vol}(S^2) = 4\pi$ and $\pi^2 = \mathrm{Vol}_1(\mathrm{SU}(2))/2$, so that $4/\pi = 2\mathrm{Vol}(S^2)/\mathrm{Vol}_1(\mathrm{SU}(2))$ in the dual-Coxeter normalisation; on the unit S^3 the constant is the plain Haar ratio $\mathrm{Vol}(S^2)/\mathrm{Vol}(\mathrm{SU}(2)) = 2/\pi$, the factor 2 being the metric scale (Paper 38). The classical Fejér kernel on $\mathbb{T}^1$ has probability-normalised first-moment constant $2/\pi$ — half the SU(2) value in the dual-Coxeter normalisation and equal to it on the unit S^3, so the factor 2 is the metric scale rather than a signature of the group (Paper 38, Remark "Connection to the circle Fejér estimate"). The $\log n$ factor in (9) is not a non-abelian slowdown: it is already present in the circle Fejér kernel (Paper 38; Paper 40), and a tapered amplitude removes it (Paper 38). Leimbach–van Suijlekom [15] prove Gromov–Hausdorff convergence of spectral truncations for tori without stating an explicit rate, so there is no published torus rate to compare against.

4.4 Lemma L3: Lipschitz comparison with $C_3 = 1$

For each $f \in C^\infty(S^3)$ and each truncated multiplier M_f on $\mathcal{O}_{n_{\max}}$, the Lipschitz comparison

$$\|[D_{\mathrm{CH}}, M_f]\|_{\mathrm{op}} \;\le\; C_3 \cdot \|\nabla f\|_{L^\infty(S^3)} \tag{10}$$

holds with $C_3 = 1$ (Paper 38, Lemma L3). The proof is by compression: $P_{n_{\max}}$ contracts operator norms, so $\|[D_{\mathrm{CH}}, M_f]\|_{\mathrm{op}} \le \|[D_{\mathrm{CH}}, M_f]\|_{\mathrm{op}} = \|\nabla f\|_{L^\infty}$ by Connes' formula on a closed spin manifold, and $C_3 = 1$ cannot be lowered uniformly because $P_{n_{\max}} \to 1$ strongly. (An earlier version of this synthesis sketched a proof through SO(4) selection rules with a per-harmonic ratio $(N-1)/\sqrt{N^2-1}$ at finite $n_{\max}$; that inequality is false and the reading was withdrawn in Paper 38, 2026-09-28.)

The same compression argument proves the bound with $C_3(G) = 1$ on every compact connected Lie group in Paper 40's class, at every rank (Paper 40 Lemma L3 [33]; its per-summand Dirac-triangle / PRV analysis concerns representation-theoretic inequalities, proved on special summands and verified on panels for the rest, whose bridge to the commutator norm is withdrawn, so it is not a check); see §5 below.

4.5 Lemma L4: Berezin reconstruction

A Berezin map $B_{n_{\max}} : C(S^3) \to \mathcal{O}_{n_{\max}}$ is constructed in Paper 38 §5.3 as

$$B_{n_{\max}}(f) \;=\; \sum_{N \le 2n_{\max}-1,\, L,\, M} \sigma_{n_{\max}}(N)\, c_{NLM}\, M_{NLM} \;=\; P_{n_{\max}}\, M_{K_{n_{\max}} * f}\, P_{n_{\max}}, \tag{11}$$

with $\sigma_{n_{\max}}$ the multiplier symbol of L2 (a sum weighted by the mass distribution $N/Z_{n_{\max}}$ instead would not be unital, Paper 38). This is the SU(2) non-Kähler analog of the Connes–van Suijlekom [6] Fejér kernel reconstruction on S^1 and the non-holomorphic analog of Hawkins [10]. Four properties are established:

(a) Positivity: for $f \geq 0$, $B_{n_{\max}}(f)$ is a positive operator on $P_{n_{\max}}\mathcal{H}_{\mathrm{GV}}$ (via convolution + positivity-preserving compression);

(b) Contractivity: $\|B_{n_{\max}}(f)\|_{\mathrm{op}} \leq \|f\|_\infty$;

(c) Approximate identity: $\|B_{n_{\max}}(f) - P_{n_{\max}} M_f P_{n_{\max}}\|_{\mathrm{op}} \leq \gamma_{n_{\max}} \|\nabla f\|_\infty$ with $\gamma_{n_{\max}}$ as in L2;

(d) L3 compatibility: $\|[D_{\mathrm{CH}}, B_{n_{\max}}(f)]\|_{\mathrm{op}} \leq \|\nabla f\|_\infty$ ($C_3 = 1$ is inherited, since $B_{n_{\max}}(f)$ is the compression of the multiplier by $K_{n_{\max}} * f$ and convolution by a probability kernel does not increase $\|\nabla f\|_\infty$).

This is the operator-system analog of the Berezin–Toeplitz quantisation; see Connes–van Suijlekom [6] for the operator-system truncation of the closely related $\mathbb{T}^d$ case.

4.6 What carries the theorem, and the withdrawn Lemma L5

The live argument: lifted states. Theorem 4.1 is Paper 38's `thm:main_unconditional`, and its proof is the *lifted-state* construction, not a propinquity assembly. Let $S : C(S^3) \to \mathcal{O}_{n_{\max}}$, $S(f) = P_{n_{\max}} M_f P_{n_{\max}}$ be the unital completely positive compression and υ the dual map of Paper 38's lifted-state lemma. Four facts close it: S and υ are each 1-Lipschitz for the Monge–Kantorovich metrics; $\|\upsilon(S(f)) - f\|_\infty \leq \gamma_{n_{\max}} L(f)$ by the good-kernel estimate; and $\|S(\upsilon(T)) - T\|_{\mathrm{op}} \leq \gamma_{n_{\max}} L_{n_{\max}}(T)$, because $S \circ \upsilon$ is exactly the operator-side conjugation average $\Phi(T) = \int \rho_g(T)\, K_{n_{\max}}(g)\, dg$ against the same Fejér kernel (7). The two maps are therefore $\gamma_{n_{\max}}$-almost mutually inverse, and the state-space pullbacks give the bound directly. *The argument uses no Berezin map and no partial inverse, so it needs no dual reach and no height term at all*—which is why the refutation below leaves it untouched.

Lemma L5, withdrawn 2026-09-03. The original architecture closed the forward direction differently, by assembling a Latrémolière tunneling pair $(B_{n_{\max}}, P_{n_{\max}})$ [13] and bounding its reach and height. That lemma is **[REFUTED AS STATED]** and withdrawn: its height leg asserted $\mathrm{height}_B \leq \gamma_{n_{\max}}$ and $\mathrm{height}_P = 0$, and a finite-band witness exhibits an f with $B(f) = 0$ for which the height quantity is exactly 1. Hence $\mathrm{height}_B \equiv 1$ at every cutoff and the bound fails (the earlier panel-height check already crossed over between $n_{\max} = 6$ and 7; monotonicity of $\gamma_{n_{\max}}$ beyond the measured range is not proved). Paper 38 retains the lemma as a record of the abandoned route; the theorem's bound has constant 1 from the lifted-state route, with no C_3 factor.

Net effect: one proof path fell and the keystone stands. WH1 is unaffected, because the unconditional theorem never depended on the height estimate. The same (B, P)-pair route is the one described as *abandoned* in §4.8 below; Paper 39's tensor extension likewise runs on the lifted-state construction.

4.7 Numerical verification

Paper 38 records the explicit values of the bound $\gamma_{n_{\max}}$ at small cutoffs (values of the bound, not computed distances):

$$\gamma_2 \approx 2.0746, \quad \gamma_3 \approx 1.6101, \quad \gamma_4 \approx 1.3223, \tag{12}$$

with $\gamma_4/\gamma_2 = 0.637$. These are monotone decreasing on this range and consistent with the $O(\log n_{\max}/n_{\max})$ Fejér-moment asymptote.

4.8 Tensor extension (Paper 39)

Paper 39 extends Theorem 4.1 to tensor products $\mathcal{T}_{S^3}^{\lambda_a} \otimes \mathcal{T}_{S^3}^{\lambda_b}$ of two copies of the same manifold at distinct focal lengths $\lambda_a \neq \lambda_b$ [32]. The result is an **[INTERNAL THEOREM]** in the translation-seminorm metrization (Paper 38's footing): convergence in van Suijlekom state-space Gromov–Hausdorff distance with bound $\Lambda \leq \gamma^{(ab)} \leq \lambda_a \gamma_{n_a} + \lambda_b \gamma_{n_b}$, *leading constant* 1, for the spinor (Camporesi–Higuchi) tensor triple with an explicit bound and the KO-dimension-6 real structure of its limit (the scalar state-space case of $\mathrm{SU}(2) \times \mathrm{SU}(2)$ is covered by Gaudillot-Estrada and van Suijlekom; the cross-manifold case stays open, Paper 32).

The proof is the *lifted-state* construction: Paper 38's `thm:main_unconditional` facts (i)–(iv), lifted factor-wise to $\mathrm{SU}(2) \times \mathrm{SU}(2)$ and metrized by the translation seminorm, give a correspondence of distortion $\leq 2\gamma^{(ab)}$; it uses neither a Berezin map nor any partial inverse, so reach_P is *unnecessary*. The limit is identified (Paper 39 Lemma real-structure-T) as the graded (Clifford-doubled) tensor product of the two odd (KO-3) Camporesi–Higuchi real spectral triples, with real structure $J_{a,b} = J_a \otimes \sigma_1 K \otimes J_b$ at KO-dimension 6; the naive $D_{\mathrm{CH}}^{(a)} \otimes I + \gamma_a \otimes D_{\mathrm{CH}}^{(b)}$ is unrealisable ($\gamma_a = 0$ on S^3) and the doubling is its replacement.

The joint Lipschitz comparison constant $C_3^{(2)} = 1$ (by compression on the Clifford-doubled module, Paper 39 Lemma L3-T) and the Berezin-tunneling cross-term (a triangle bound, rate $O(\max \gamma)$) belong to an *abandoned* (B, P)-pair route, retained in Paper 39 as a record; they are *not* rate constants of the theorem, whose constant is 1. (That route's earlier Pythagorean refinement $C_3^{(2)} < 1$ was withdrawn as false; the triangle bound on the two Leibniz terms is tight term by term and gives only $\sqrt{2}$.)

5 The Fejér-moment constant beyond SU(2)

Paper 40 [33] extends Theorem 4.1 from SU(2) to compact connected semisimple Lie groups with bi-invariant metric (for general G conditionally on the per-group spin-window decomposition).

Theorem 5.1 (Paper 40, compact Lie groups)**.** *Let G be a compact connected semisimple Lie group of rank $r \geq 1$ with bi-invariant metric (each simple factor $\in \{\mathrm{SU}(N), \mathrm{Spin}(n)\ (n \geq 5), \mathrm{Sp}(n), G_2, F_4, E_6, E_7, E_8\}$). The truncated (Kostant-Dirac) triples $\mathcal{T}_{n_{\max}}(G)$ converge to $\mathcal{T}_G$ in the van Suijlekom state-space Gromov–Hausdorff sense; the Fejér-type moment's leading constant is that of (6) at rank 1, and whether it is the same for every simple G is **[OPEN]**: the rank-2 extractions used a normalisation that does not reproduce the* SU(2) *anchor (Paper 40; for products of several simple factors the constant is also open); whether a tapered amplitude gives $O(1/n_{\max})$ for general G is **[OPEN]**. The Lipschitz comparison constant is $C_3(G) = 1$ at every rank, by compression. For general G the convergence is conditional on a per-group spin-window decomposition (verified for* SU(2)*).*

Plancherel-weight × Vandermonde-Jacobian cancellation. Paper 40 proposes a mechanism for a group-independent constant (a proof sketch; whether the constant is in fact group-independent is open, Theorem 5.1). In the central spectral Fejér kernel on G,

$$K_{n_{\max}}(g) \;=\; \frac{1}{Z_{n_{\max}}(G)} \left| \sum_{\lambda\,:\,|\lambda+\rho|^2 \leq M(n_{\max})^2} \sqrt{\dim V_\lambda}\, \chi_\lambda(g) \right|^2 , \tag{13}$$

the Plancherel weight $\sqrt{\dim V_\lambda}$ on the squared amplitude (so the weight is $\dim V_\lambda$) and the Vandermonde Jacobian $|\Delta(t)|^2$ in the Weyl integration formula on G share the same $\prod_{\alpha>0}$ product structure over positive roots. The sketch has these products cancel, leaving the rank-1 (SU(2)) constant of Paper 38 Appendix A from each positive-root direction's Euler sum $\sum_{d\,\mathrm{odd}} 1/d^2 = \pi^2/8$.

The PRV-summand inequalities (representation theory, not a Lipschitz bound). The Lipschitz comparison bound (10) holds on G by compression (Paper 40 Lemma L3). Separately, Paper 40 studies representation-theoretic inequalities on the PRV summands (existence by Kumar 1988; an earlier attribution to Vinberg 1990 carried no locator and is withdrawn, Paper 38): for every Weyl group element $w \in W$, the dominant image σ_w of $\lambda + w\lambda'^*$ is a summand of $V_\lambda \otimes V_{\lambda'}^*$, and Paper 40's Theorem (b) gives $|\sigma_w + \rho|^2 \geq |\lambda - \lambda'|^2$, which falls short of the intermediate inequality (INT) by $|\rho|^2$, so on the PRV class (INT) rests on the exact-arithmetic panels. Interior (non-PRV) summands are verified across $\mathrm{SU}(3), \mathrm{SU}(4), \mathrm{Sp}(2), G_2$ at 5641/5641 cells. These are statements about eigenvalues and Casimirs; earlier versions bridged them to the commutator norm (and read the eigenvalue ratio as a per-isotypic Lipschitz constant), the step Paper 38 refuted at rank 1, so they are not a check of C_3.

Companion: state-space GH. The state-space Gromov–Hausdorff convergence (without rate and without Dirac) at the same generality was proved by Gaudillot-Estrada–van Suijlekom (IMRN 2025, paper rnaf197). Paper 40 provides the Fejér-moment bound and its constant, the Lipschitz spinor bound, and the asymptotic tightness on the state-space GH substrate; the further step from state-space GH to spectral-triple-level Latrémolière propinquity (the dual-reach step) is a conceded named gap (the same scope boundary as Paper 38). Separately, for general G the result is conditional on a per-group spin-window decomposition, verified only for SU(2).

6 Modular structure and Wick rotation

6.1 The four-witness 2π period at finite cutoff (Paper 42)

The Wick rotation between Euclidean and Lorentzian field theory is classically formulated through four equivalent presentations on a flat spacetime:

- Hartle–Hawking 1976 [9]: the temperature of the Euclidean cigar's τ-circumference identifies as the Hawking temperature;
- Sewell 1982 [23]: the KMS state at inverse-temperature $\beta = 2\pi/\kappa_g$ is the wedge vacuum;
- Bisognano–Wichmann 1976 [1]: the modular automorphism group of the wedge von Neumann algebra acts as boosts around the wedge edge;
- Unruh 1976 [26]: an accelerated observer with proper acceleration a sees a $T_U = a/(2\pi)$ thermal bath in the vacuum.

On a flat background, the four are equivalent under standard Wick-rotation arguments. On a curved background like S^3 they are not obviously equivalent, and at finite cutoff (operator-system level) they require separate construction.

Paper 42 [34] does *not* establish the four-witness theorem at the operator-system level (an earlier reading that it did, at every finite cutoff, is withdrawn, 2026-09-28). It sets up the finite-cutoff ingredients on the truncated Camporesi–Higuchi spectral triple $\mathcal{T}_{n_{\max}}$ with a *posited* generator, and records two readings of one flow — conjugate by construction, not independent. Every closure below is elementary [SYMBOLIC]: the period closure holds for *any* integer-spectrum generator and any operator, the Tomita step returns the generator the Gibbs state was built from, and the witnesses agree because they share one generator with κ_g relabelled. The continuum four-witness statement is prior art; deriving the wedge modular Hamiltonian from a Camporesi–Higuchi state is open (Paper 42, O0).

Theorem 6.1 (Paper 42, BW-α period closure; [SYMBOLIC], automatic for integer spectrum)**.** *For each $n_{\max} \in \{2, 3, 4, 5\}$, there exists a wedge projection $P_W = \frac{1}{2}(I + R_{\mathrm{polar}})$ on $P_{n_{\max}}\mathcal{H}_{\mathrm{GV}}$*

(bit-exact idempotent and Hermitian), and for the posited geometric generator

$$K_\alpha^W \;=\; \mathrm{diag}(2m_j) \text{ on } P_W\mathcal{H}_{\mathrm{GV}} \quad \textit{(a relabelling; Paper 42, Def. K_alpha_W)}, \tag{14}$$

with integer spectrum $\{2m_j\}$ on the full-Dirac basis. The conjugation flow it generates (a modular flow only if Paper 42's open O0 identifies it) satisfies $\sigma_{2\pi}(O) = O$ bit-exact (residual $O(\sqrt{\dim\mathcal{H}}\cdot\varepsilon_{\mathrm{machine}})$) for the four witness instantiations BW, Hartle–Hawking, Sewell, Unruh, with cross-witness residual identically zero (by construction: one generator, κ_g relabelled).

Theorem 6.2 (Paper 42, BW-γ Tomita–Takesaki reading; [SYMBOLIC], Tomita's theorem for a Gibbs state)**.** *On the Krein-GNS Hilbert–Schmidt space $\mathcal{H}_{\mathrm{GNS}} = M_{\dim W}(\mathbb{C})$ with KMS state $\rho_W = e^{-K_\alpha^W}/Z$ and the BW choice $H_{\mathrm{local}} := K_\alpha^W/\beta$, the polar decomposition $S = J_{\mathrm{TT}}\cdot\Delta^{1/2}$ gives a Tomita-Takesaki modular Hamiltonian $K_{\mathrm{TT}} = -\log\Delta$ with the same integer spectrum, and the operator-action conjugacy*

$$\sigma_t^{\mathrm{TT}}(a) \;=\; \sigma_{-t}^{\alpha}(a) \tag{15}$$

holds bit-exact at every tested $t \in \{1, \pi, 2\pi\}$. The modular operator Δ satisfies $J_{\mathrm{TT}}^2 = +I$, categorically distinct from the Connes J_{GV} (KO-dim 3, $J^2 = -I$).

Operator-level contrast (Paper 42 §7.2). The local Hamiltonian $H_{\mathrm{local}} = K_\alpha^W/\beta$ is a posit, not derived; it is contrasted with the intrinsic Camporesi–Higuchi Dirac operator D_{CH}. The intrinsic D_{CH} has half-integer spectrum, so its operator-level period lift is $e^{i\cdot 2\pi\cdot D_{\mathrm{CH}}} = -I$ (the spinor double-cover sign), whereas the odd-integer K_α^W lifts to $+I$. The *conjugation* modular flow $\sigma_{2\pi}$ closes with either generator (a global scalar cancels in the conjugation), so the distinction is at the operator level, not the flow-closure level: only K_α^W lifts the period to the identity operator. Which generator (if either) a Camporesi–Higuchi wedge state actually selects is open (Paper 42, O0). This is the open question O3 of Paper 42 §10, sharpened in Paper 42 §11.6 to the signature-independence statement below.

6.2 Lorentzian extension at finite cutoff (Paper 42 §11)

Paper 42 §11 [34] (which absorbs the former Paper 43, archived 2026-09-28) carries the same construction to the Lorentzian Krein space at signature $(3,1)$ (automatic, and signature-blind on the argued reading [ARGUED]; not a finite-cutoff four-witness theorem), built via the Bizi–Brouder–Besnard prescription [2] at $(m,n) = (4,6)$ in the Peskin–Schroeder chiral basis, with Lorentzian Dirac

$$D_L \;=\; i\left(\gamma^0 \otimes \partial_t \;+\; D_{\mathrm{GV}} \otimes I_{N_t}\right) \tag{16}$$

per van den Dungen 2016 Proposition 4.1 [27].

Theorem 6.3 (Paper 42, Theorem 11.9, Krein-carrier period closure at finite cutoff; [SYMBOLIC], automatic)**.** *At every tested $(n_{\max}, N_t) \in \{1,2,3\}\times\{1,11,21\}$, the Krein-level analogues of (14) and (15) hold bit-exact:*

- *(i) The posited Lorentzian wedge generator $K_L^{\alpha,W}$, the relabelling $\mathrm{diag}(2m_j)$ on W_L (its Bisognano–Wichmann identification is open, Paper 42 O0) has integer spectrum on the Krein hemispheric wedge.*
- *(ii) The Krein-Tomita modular operator Δ_L admits a polar decomposition $S = J_{L,\mathrm{TT}}\cdot\Delta_L^{1/2}$; the Tomita modular Hamiltonian K_L^{TT} has integer spectrum.*
- *(iii) $\sigma_{2\pi}^{L,\alpha}(O) = O$ and $\sigma_{2\pi}^{L,\mathrm{TT}}(a) = a$ bit-exact, with cross-witness collapse residual zero by construction (the witnesses share one generator, differing only in κ_g, which cancels).*
- *(iv) At $N_t = 1$, the Lorentzian construction reduces to the Riemannian one of Theorems 6.1–6.2 bit-identically.*

Signature-independence of the operator-level contrast. A finding of Paper 42 §11.6 is that the "$H_{\text{local}} \neq D_W^L$" statement is *signature-independent* at the Riemannian reduction. At $N_t = 1$ the residual norm $\|H_{\text{local}} - D_W^L\|_F$ on the Lorentzian Krein wedge equals the Riemannian-side value at every $n_{\max}$ (Paper 42, Theorem 11.11):

$$\|H_{\text{local}}|_{(3,1),N_t=1} - D_L^W|_{N_t=1}\|_F \;=\; \|H_{\text{local}}^{\text{Riem}} - D_W\|_F \qquad (\text{at every } n_{\max}).$$

At $N_t > 1$ the Lorentzian residual is refined upward by the temporal-derivative content of D_L: the spectral-action object D_L has more content than D_{GV}, so the gap between H_{local} (the posited generator; its modular-Hamiltonian identification is open, Paper 42 O0) and D_W^L (spectral-action object) widens with temporal cutoff. This refines the Paper 42 §10 open question O3 from "open about signature-dependence" to a signature-independent property of the posited construction (not evidence about which generator a Camporesi–Higuchi wedge state selects).

6.3 Pythagorean HS-orthogonality and the $1/\pi^2$ signature

Paper 42 §11.7 records a Pythagorean refinement of the modular vs spectral-action gap. Let $\langle \cdot, \cdot \rangle_{\text{HS}}$ denote the Hilbert–Schmidt inner product on the wedge operator algebra.

Corollary 6.4 (Paper 42, Theorem 11.14 and Corollary 11.15, HS-orthogonality on the Lorentzian wedge)**.** *For every $\kappa_g > 0$ and every panel cell $n_{\max} \in \{1, \ldots, 6\}$, $N_t \in \{1, 11\}$, and witness $w \in \{\text{BW}, \text{HH}_1, \text{HH}_2, \text{Sew}, \text{Unruh}_1, \text{Unruh}_2\}$:*

$$\langle H_{\text{local}}, D_W^L \rangle_{\text{HS}} \;=\; 0 \qquad (\textit{bit-exact}), \tag{17}$$

and, at $N_t = 1$ only (Paper 42 Corollary 11.15; the orthogonality (17) holds on every cell), the residual norm has the closed form

$$\|H_{\text{local}} - D_W^L\|_F^2\big|_{N_t=1} \;=\; \frac{\kappa_g^2}{4\pi^2}\, S(n) + D(n), \quad S(n) = \frac{n(n+1)(n+2)(2n^2+4n-1)}{15}, \quad D(n) = \frac{n(n+1)(n+2} \tag{18}$$

The closed form (18) was identified by PSLQ at 100 digits (coefficient ceiling 10^6) at $n_{\max} = 1, \ldots, 6$ in the original measurement; that PSLQ step is not in the regression suite. The backing test recomputes S and D from the live wedge operators at $n_{\max} = 1, \ldots, 4$ and checks the orthogonality on 7 cells (Paper 42 Corollary 11.15). The $1/\pi^2$ prefactor on $\|H_{\text{local}}\|^2$ enters through $H_{\text{local}} = \kappa_g K_\alpha^W/(2\pi)$, i.e. through the posited $\beta = 2\pi/\kappa_g$ — the continuum KMS period imported as a normalisation, an input rather than a computed output. It lies in the algebraic class $\pi^{-2}\mathbb{Q}$ of the M_1 ($k = 0$ volume-slot) sector, but no Mellin mechanism producing it is shown and no identity with the M_1 content of the Paper 38 Fejér-moment constant $4/\pi$ (9) is established (see §8).

Chirality-parity mechanism. The mechanism is a wedge-chirality parity argument: $H_{\text{local}} = K_\alpha^W/\beta$ (eigenvalues two m_j, chirality-independent) is even under the wedge chirality grading, while D_W^L has a chirality-odd temporal factor and a chirality-even but sign-paired spatial factor, so both HS inner products vanish. The formal proof is Paper 42's Theorem 11.14 (§9.3 below); it is β-independent, so the $1/\pi^2$ plays no role in it. The closed form (18), stated at $N_t = 1$, is [MEASURED]: PSLQ-identified at $n_{\max} \leq 6$ and test-recomputed at $n_{\max} \leq 4$.

7 Lorentzian propinquity

7.1 Operator-system substrate (Paper 44)

The first step in any Lorentzian propinquity proof is an operator- system substrate. Paper 44 [35] provides it on the chirality-doubled Camporesi–Higuchi spinor space $\mathcal{K}_{n_{\max},N_t} = \mathcal{H}_{\text{GV}}^{n_{\max}} \otimes \mathbb{C}^{N_t}$ at BBB signature $(m, n) = (4, 6)$.

Theorem 7.1 (Paper 44, propagation number and envelope dependence). *The truncated Lorentzian operator system* $\mathcal{O}^L_{n_{\max},N_t}$ *satisfies*

$$\mathrm{prop}(\mathcal{O}^L_{n_{\max},N_t}) = 2 \quad \textit{under the Weyl-doubled achievable envelope}, \quad \mathrm{prop}(\mathcal{O}^L_{n_{\max},N_t}) = \infty \quad \textit{under the full (di} \tag{19}$$

generalising the Riemannian prop = 2 *result for* $\mathcal{O}_{n_{\max}}$ *(Paper 32 §3.4) to the Lorentzian Krein setting. The witness pair* $(M^{2,1,0,0}, M^{2,1,0,0*})$ *at* $n_{\max} = 2$ *exhibits* $\sim 38\%$ *Frobenius residual against* $\mathcal{O}^L_{2,3}$. *Riemannian-limit recovery at* $N_t = 1$ *is bit-exact (Frobenius residual* $= 0.0$ *in float64) across* $n_{\max} \in \{1, 2, 3\}$.

K^+-positivity is trivial at the operator-multiplier level. A key structural observation of Paper 44 is that all chirality- doubled scalar multipliers $M \oplus M$ commute with $J_L =$ chirality-swap, and therefore preserve $\mathcal{K}^+$ automatically: $\mathcal{O}^{L,+} = \mathcal{O}^L$ exactly. Consequently the non-trivial Krein-positivity program shifts from operator level to *state* level (Wasserstein–Kantorovich on K^+ states), the setting in which Paper 45's degeneracy theorem (below) is read.

7.2 K^+-restricted weak-form Lorentzian propinquity: the degeneracy theorem (Paper 45)

Paper 45 [36] closes off the Lorentzian-propinquity route via a degeneracy theorem (the natural K^+-weak-form seminorm vanishes identically) on the substrate of Paper 44.

Definition 7.2 (Paper 45 §1.3, K^+-weak-form Lorentzian propinquity). For two truncated Lorentzian spectral triples $\mathcal{T}^L_i$ $(i = 1, 2)$, define

$$\Lambda^L(\mathcal{T}^L_1, \mathcal{T}^L_2) := \Lambda(\mathcal{T}^L_1|_{\mathcal{K}^+}, \mathcal{T}^L_2|_{\mathcal{K}^+}), \tag{20}$$

where Λ is the quantum Gromov–Hausdorff-type distance the device was meant to supply (Paper 45 frames it in van Suijlekom's state-space setting, Latrémolière's tunnel hypotheses [13, 14] being unavailable there) and the restriction to $\mathcal{K}^+ = \{|\psi\rangle : J_L|\psi\rangle = +|\psi\rangle\}$ is the standard metric spectral triple obtained from the Krein-positive subspace, on which the indefinite Krein product reduces to a positive-definite Hilbert inner product.

Theorem 7.3 (Paper 45, annihilation theorem). *The device of Definition 7.2 is degenerate. For* D_L *as in (16), Krein-self-adjointness of the anti-Hermitian spatial block forces* $\{J_L, D_{\mathrm{GV}} \otimes I\} = 0$, *hence* $P^+(D_{\mathrm{GV}} \otimes I)P^+ = 0$*: the compression annihilates the spatial Dirac. The surviving temporal block is momentum-diagonal and commutes with every momentum-diagonal temporal multiplier, so the restricted Lipschitz seminorm vanishes identically on the operator system and* Λ^L *of (20) is not a quantum Gromov–Hausdorff-type distance. (Bit-exact: restricted-seminorm kernel* 42/42 *and* 275/275 *multipliers at* $(n_{\max}, N_t) = (2, 3), (3, 5)$*; frozen falsifier in the repository test suite.)*

An earlier convergence claim through this device (formerly Theorem 5.1 of Paper 45) is withdrawn. The rate expression

$$C_3^{\mathrm{joint}} \cdot \gamma^{\mathrm{joint}}_{n_{\max},N_t,T} = O\big(\log n_{\max}/n_{\max} + T \log N_t/N_t\big) \tag{21}$$

survives as a well-defined closed form; its spatial leg $\gamma^{\mathrm{SU}(2)}$ is the bound of the unconditional *spatial* statement (Paper 38, constant 1), the temporal factor being a metrically invisible spectator.

7.3 Three structural simplifications

The original transfer of the Paper 38 lemma chain (five lemmas as then stated; the fifth withdrawn 2026-09-03) to the joint SU(2) × U(1) setting recorded three structural "simplifications." Under the degeneracy analysis each is reinterpreted: they are *symptoms* of the temporal factor's metric invisibility, not conveniences:

PURE_TENSOR Berezin map. The joint Berezin map $B^{\text{joint}} = B^{\text{SU}(2)} \otimes B^{\text{U}(1)}$ factorises as a pure tensor. Each of the four required properties (positivity, contractivity, approximate identity, L3 compatibility) inherits factor-by-factor; no cross-factor verification is needed.

Vanishing time-chirality cross-term. The joint Lichnerowicz commutator simplifies via the structural identity

$$[D_L,\, a_s \otimes a_t] \;=\; i\,[D_{\text{GV}}, a_s] \otimes a_t \tag{22}$$

because $[\gamma^0, a_s] = 0$ (the spatial multiplier is chirality-blind) and $[\partial_t, a_t] = 0$ (the temporal algebra is momentum-diagonal), so the $\gamma^0 \otimes \partial_t$ term contributes nothing (Paper 45, Lemma L3). The temporal direction contributes nothing to the joint commutator, and $C_3^{\text{joint}} = C_3^{\text{SU}(2)}$ verbatim. Read correctly, (22) is the *diagnosis*: the momentum-diagonal temporal algebra is invisible to the Dirac commutator.

Corrected cb-norm convention. The smoothing map of the joint central spectral Fejér kernel is unital completely positive, with cb-norm exactly 1; the quantity $2/(n_{\max}+1)$ formerly quoted as the "joint cb-norm" is the maximum of the SU(2) Plancherel *mass distribution*, a different invariant (the Bożejko–Fendler appeal is withdrawn; the correct statement is elementary). The N_t-dependence of the rate (21) lives entirely in the joint mass-concentration moment (the $T \log N_t / N_t$ Fejér-on-the-circle rate).

Riemannian-limit recovery at $N_t = 1$ bit-exact. At $N_t = 1$, the joint Berezin reduces bit-identically to $B^{\text{SU}(2)} \otimes B^{\text{U}(1)}_{1,T} = B^{\text{SU}(2)}$, since the $N_t = 1$ temporal Berezin map is the identity on the single Fourier mode. Hence the spatial rate $\gamma^{\text{SU}(2)}_{n_{\max}}$ computed by the construction at $N_t = 1$ agrees with Paper 38's bit-exactly. Under the degeneracy analysis this recovery is an automatic formula identity (the panel quantity only ever contained the spatial rate formula), not an operator-level falsifier; the falsifier with teeth is the restricted-seminorm check of Theorem 7.3.

7.4 Honest scope

Theorem 7.3 closes the natural device; the Lorentzian convergence question itself remains open *at the continuum / non-compact level.* The *finite-cutoff* metric question has a structural argument (Paper 45 §open Q1, 2026-06; re-tiered 2026-09-28): every finite-cutoff generator, posited or derived, generates a flow with compact closure, never a non-compact boost [SYMBOLIC]. That a compact flow carries only a Euclidean (forward-triangle) structure — which is why the forward chain inequality is the only universal direction and every Lipschitz / boost seminorm is signature-blind — is [ARGUED], so the finite-cutoff quantum metric is Euclidean independently of O0 but on an argued, not proved, step. For Paper 42's *posited* generator the spectrum is integer ($e^{2\pi i K} = I$), so its flow is a 2π-periodic circle; reading that circle as the KMS circle of the Bisognano–Wichmann boost at finite cutoff, and the Lorentzian signature as its Wick rotation $t \to it$ (a convention rather than a finite-cutoff metric), is conditional on Paper 42's open O0. The genuine reverse-triangle Lorentzian signature requires the non-compact continuum ($n_{\max} \to \infty$, type-III) limit, which is the sense in which the question remains open. This is the metric-level dual of the WH7 reading (compactness ⇒ discreteness ⇒ π/β). Several aspects of scope require explicit comment.

(Q1) **Strong-form Lorentzian propinquity (Paper 45 §1.4 G1).** The K^+-restriction reduces the indefinite Krein product to a Hilbert inner product on $\mathcal{K}^+$ — but annihilates the spatial Dirac (Theorem 7.3), so no quantum-metric machinery applies on the compression. Strong-form Lorentzian propinquity (Latrémolière-style metric on Krein spectral triples in their own right, without K^+-restriction) requires an indefinite-inner-product analog of the operator-norm Lipschitz seminorm with appropriate behaviour under Krein-self-adjoint D_L. This remains a substantial open NCG-math problem.

(G2) **De-compactification $T \to \infty$.** The temporal factor is compactified to $U(1)_T$ with finite radius T. De-compactification to non-compact $\mathbb{R}_t$ is a separate program (internal Sprint L3c) requiring renormalisation of the mass-concentration moment and reach moment.

(G3) **Cross-manifold $\mathcal{T}_{S^3} \otimes \mathcal{T}_{\mathrm{Hardy}(S^5)}$.** Blocked at the NCG-framework level by the Riemannian-vs-Hardy-sector asymmetry recorded in Paper 24 §V of the GeoVac corpus (the Coulomb/HO asymmetry: seven layers catalogued, five standing); this is the Coulomb-side claim only.

(G4) **Inner-factor calibration data.** Higgs / Yukawa selection question of the Connes–Marcolli SM construction; orthogonal to Paper 45's results (a degeneracy theorem plus the surviving spatial statement), addressed by Paper 32 §8.C verdict POSITIVE-THIN (construction admits Higgs structurally, but no autonomous Yukawa selection from the present data).

8 The master Mellin engine and the cross-arc connection

The most striking structural observation that emerges from reading the arc is that transcendental signatures of the same algebraic class recur across settings. The Fejér-moment constant $4/\pi$ of Paper 38 is read (a reading, not a derivation) as the $k = 0$ volume content, the M_1 sub-case of the master Mellin engine catalogued in Paper 32 §8. The constant $1/\pi^2$ of Paper 42 §11.7 lies in the same ring $\pi^{-2}\mathbb{Q}$ but is *not* shown to arise from that mechanism: it is the square of the posited normalisation $\beta = 2\pi/\kappa_g$.

8.1 The π-source case-exhaustion theorem (Paper 32 §8)

Theorem 8.1 (Paper 32 §8, π-source case-exhaustion)**.** ***[CONDITIONAL]*** *Let $\mathcal{C}$ be any finite composition of the first fifteen Paper 34 projections applied to a Layer-1 datum on $(\mathcal{A}_{\mathrm{GV}}, \mathcal{H}_{\mathrm{GV}}, D_{\mathrm{GV}})$ or its Bargmann–Segal sibling triple. Then (i)⇔(ii) (empirical, Paper 35) and (ii)⇒(iii) (case-by-case argument); the converse (iii)⇒(i) is not established (M3's odd-zeta values at half-integer s and its convergent value at $s = 4$ have no known π):*

(i) $\mathcal{C}$(datum) contains a transcendental of the form π^k for $k \in \mathbb{Q}$;

(ii) $\mathcal{C}$ contains at least one projection whose evaluation includes a continuous integration over a parameter promoted from the discrete graph spectrum;

(iii) $\mathcal{C}$ engages at least one of three spectral-triple mechanisms,

- M_1 *($k = 0$ volume slot): $\pi = \mathrm{Vol}(S^2)/4$, a quarter of the unit-sphere area $\mathrm{Vol}(S^2) = 4\pi$ (equivalently the full area of the Hopf base $S^2(1/2)$ of the unit-S^3 Hopf bundle $S^1 \to S^3 \to S^2$); the $4/\pi$ Fejér-moment constant is twice the unit-sphere volume quotient $\mathrm{Vol}(S^2)/\mathrm{Vol}(S^3) = 2/\pi$ (convention-dependent: $4/\pi$ dual-Coxeter, $2/\pi$ on the unit S^3);*
- M_2 *(*Seeley–DeWitt*): $\sqrt{\pi}$ via heat-kernel coefficients on the Camporesi–Higuchi Dirac spectrum on S^3 (Paper 28 Theorem T9: $\zeta_{D^2}(s)$ is polynomial in π^2 with rational coefficients at every integer s, odd zeta at half-integer $s \geq 5/2$);*
- M_3 *(*vertex-topology Dirichlet L*): Catalan G and $\beta(s)$ via quarter-integer Hurwitz shifts when the Dirac spectrum is restricted to one parity class (Paper 28 Theorem 3).*

The TS-E3 sharpening of Paper 32 further reads M_1, M_2, M_3 as sub-cases of a single master Mellin-engine mechanism

$$\pi\text{-source} \;=\; \mathcal{M}\big[\mathrm{Tr}(D^k e^{-tD^2})\big] \quad \text{with } k \in \{0,1,2\}, \tag{23}$$

where $k=0$ selects M_1, $k=1$ selects M_3 (read through the absolute sum and its vertex-parity grading, since the plain trace vanishes on the symmetric spectrum; Paper 18), and $k=2$ selects M_2.

8.2 M_1 in the Fejér-moment constant

The Fejér-moment constant $4/\pi$ in (6) carries the M_1 ($k=0$ volume) signature, a convention-dependent reading; it says nothing about the GH rate, which a tapered amplitude makes $O(1/n_{\max})$ (Paper 38). The value's universality across the class follows from the Plancherel-weight × Vandermonde-Jacobian cancellation mechanism (Paper 40 §3.2; derived at rank 1, proof-sketch tier at higher rank) which leaves only the rank-1 (SU(2)) constant after the $\prod_{\alpha>0}$ products cancel. Equivalently, the single-positive-root Euler–Catalan sum

$$\sum_{d \text{ odd}} \frac{1}{d^2} \;=\; \frac{\pi^2}{8}$$

produces the π^2 in the denominator, and the unit-sphere area $\mathrm{Vol}(S^2) = 4\pi$ supplies the numerator ($4/\pi$ is twice the unit-sphere volume quotient $\mathrm{Vol}(S^2)/\mathrm{Vol}(S^3)$, not a Hopf-base quantity: the Hopf base $S^2(1/2)$ of the unit S^3 has area π).

8.3 The $1/\pi^2$ of the Lorentzian HS-orthogonality (re-tiered)

The $1/\pi^2$ prefactor on $\|H_{\mathrm{local}}\|^2$ in the Pythagorean closed form (18) has a single, explicit source: the Krein wedge BW choice $H_{\mathrm{local}} = K_\alpha^W/\beta$ with the *posited* $\beta = 2\pi/\kappa_g$ introduces a factor $1/(2\pi)$ (at $\kappa_g = 1$), which squared gives $1/(4\pi^2)$. That 2π is the continuum KMS/Bisognano–Wichmann period imported as a normalisation (Paper 42; the finite-cutoff generator is posited, not derived from a state, O0 open) — a temporal-compactification π in the sense of Paper 35, entering as input. It is in the algebraic class of M_1, but it is *not* shown to be the same M_1 signature as the $4/\pi$ of Paper 38, and no $\mathrm{Vol}(S^2) = 4\pi$ identity links them.

Why this matters. The case-exhaustion theorem (Theorem 8.1) is a *spectral-triple-level* statement about where π can enter chains of the first fifteen Paper 34 projections (**[CONDITIONAL]**). The constant $4/\pi$ of Paper 38 is a *kernel-level* statement about the Fejér moment that bounds the state-space GH distance. The fact that these two abstractly-different levels agree on the same signature is a consistency check within one normalisation (in Kac's normalisation the constant is $2\sqrt{2}/\pi$, so the agreement is convention-dependent): the Fejér kernel on the GeoVac spectral triple is *read* as carrying the same M_1 signature that the abstract Mellin engine catalogues — a reading, not a derivation. (Paper 42 §11.7's Lorentzian Pythagorean decomposition does *not* extend this: its $1/\pi^2$ is the imported normalisation of the posited generator, not a computed M_1 output.)

9 Closures since drafting (May 23, 2026)

The synthesis above (§1–§8) was drafted around the arc Papers 29, 32, 38, 39, 40, 42, 43, 44, 45. On 2026-05-23, four substantive closures landed in a single session, adding Papers 46 and 47 to the arc and promoting two named open follow-ons within Papers 39 and 43. This section records the four closures in summary form.

9.1 Strong-form Lorentzian propinquity (Paper 46)

Status: the strong-form convergence claim of this subsection is withdrawn. On the natural chirality-doubled substrate the operator-norm Lipschitz seminorm L_{op} has kernel far exceeding the scalars (every temporal multiplier $\mathbf{1} \otimes M^{\mathrm{temp}}$ is L_{op}-invisible), so neither side of (24) is a quantum-metric distance between quantum compact metric spaces; the bit-exact panel match below re-evaluates the same spatial rate formula and is not an independent convergence. The enlarged-substrate algebra of Appendix B survives and is the named starting point for the repair.

Paper 45 §1.4 G1 named the strong-form Lorentzian propinquity (a Latrémolière-style metric on Krein spectral triples without K^+-restriction) as the principal open NCG-math problem of the Lorentzian arc. Paper 46 [37] *claimed to close* this question on the natural chirality-doubled scalar-multiplier substrate of Paper 44 [35] via the operator-norm Lipschitz seminorm $L_{\mathrm{op}}(a) := \|[D_L, a]\|_{\mathrm{op}}$:

$$\Lambda^{\mathrm{strong}}\big(\mathcal{T}^L_{n_{\max},N_t,T},\ \mathcal{T}^L_{\mathcal{M}}\big) \ \le\ C_3^{\mathrm{op}}(n_{\max}) \cdot \gamma^{\mathrm{joint}}_{n_{\max},N_t,T} \ \to\ 0, \tag{24}$$

with operator-norm constant C_3^{op} — for which the earlier closed form $\sqrt{1 - 1/n_{\max}}$ is *withdrawn* as operator-norm-false (the correct comparison is the Paper 38 Lemma L3 value $C_3 = 1$; see Paper 46 App. A.3) — and rate $\gamma^{\mathrm{joint}} = O(\log n_{\max}/n_{\max} + T \log N_t/N_t)$ from Paper 45. The numerical panel matches Paper 45's K^+-weak-form panel *bit-exactly* at every tested cell — the "free-upgrade" reading.

Paper 46's Appendix B (added 2026-05-22; Paper 46 is descoped) claimed to extend the closure to the *enlarged* operator system with chirality-flipping generators satisfying $\{J_L, M^{\mathrm{flip}}\} = 0$. The substantive new content is the *gradient-norm absorption mechanism*: the strict-strong-form separation between enlarged and K^+-weak-form substrates manifests at the Lipschitz seminorm level ($L_{\mathrm{op}}(a^{\mathrm{flip}}) > 0$ on enlarged generators while $L^+_{\mathrm{P45}}(a^{\mathrm{flip}}) = 0$ identically), but the separation is paid *entirely in a J-graded gradient-norm extension* $G^{\mathrm{enlarged}} = G^{\mathrm{natural}} + 2\|D_t\|_{\mathrm{op}}\|f^{\mathrm{flip}}\|_{\mathrm{op}}$; the two panel values coincide, $\Lambda^{\mathrm{enlarged}}(n_{\max}, N_t) = \Lambda^{\mathrm{P46}}(n_{\max}, N_t)$ at every cell, because both evaluate the same spatial rate formula — not because a propinquity bound is preserved (none is established).

9.2 De-compactification limit, norm-resolvent / spectral closure (Paper 47)

Status: the outer *norm-resolvent arrow of (25) and the three-carrier identification stand; the* inner *propinquity arrow (Paper 45/46 main theorem applied along the scaling) is descoped by Paper 45's degeneracy theorem, so the composite is established at the norm-resolvent / spectral level only.*

Paper 45 §1.4 G2 named the de-compactification limit $T \to \infty$ on the temporal carrier S^1_T as a separate open program. Paper 47 [38] closes it at the *norm-resolvent / spectral* level via a *two-rate hybrid composite*:

$$\mathcal{T}^L_{n_{\max},N_t,T(N_t)} \xrightarrow{\Lambda^L\text{-rate (descoped) } O(\log n_{\max}/n_{\max} + T(N_t)\log N_t/N_t)} \mathcal{T}^L_{S^3\times S^1_{T(N_t)}} \xrightarrow{\text{nr-rate } O(e^{-|\Im z|T(N_t)/2})} \mathcal{T}^L_{S^3\times\mathbb{R}_t} \tag{25}$$

along any *admissible scaling* $T(N_t) \to \infty$ with $T(N_t)\log N_t/N_t \to 0$ (the circle Fejér rate; an earlier version wrote $T(N_t)/N_t$ [retracted 2026-10-02: fejer-circle-rate-T-over-N], under which the sub-linear scaling $T_1 = N_t/\log N_t$ below looked admissible; under the corrected rate it is borderline, the rate tending to a constant). The inner arrow was proposed as propinquity convergence at coupled cells (Paper 45/46 main theorem applied along the scaling), but that arrow is *descoped* by Paper 45's degeneracy theorem; the outer arrow is norm-resolvent convergence of the Lorentzian Dirac on the compactified spacetime to the Lorentzian Dirac on the genuine spacetime, with explicit exponential tail bound $O(e^{-|\Im z|T/2})$ on compact-support subspaces via

Reed–Simon [39] § XIII.16 / Bloch-decomposition machinery and the Krein-isometric embedding $\widetilde{J}_T : L^2(S^1_T) \hookrightarrow L^2(\mathbb{R})$.

A strategic clarification appears in Paper 47 §1.1: the natural *inductive-limit propinquity* framework of Latrémolière [14] § 5 for AF C*-algebras is structurally wrong for the present setting because AF C*-algebras have totally-disconnected Gelfand spectrum while $\mathbb{R}$ is connected, so $C_0(\mathbb{R})$ cannot be an AF inductive limit. Norm-resolvent convergence is the right framework at the spectral level.

Paper 47 §6 also establishes a *three-carrier identification*: the periodic compact $S^3 \times S^1_T$ (Paper 45 substrate), the bounded Dirichlet $S^3 \times [-T/2, T/2]$ (Paper 42 §11 / Sprint L2-B/E substrate), and the non-compact $S^3 \times \mathbb{R}_t$ (Paper 47 limit) all converge in norm-resolvent sense to the same physical Lorentzian Dirac as $T \to \infty$. This resolves the apparent tension between Paper 42 §11's bounded-interval architecture (chosen to preserve causal structure / no CTCs via Geroch's theorem) and Paper 45's periodic-compact architecture (chosen, originally, to enable Latrémolière propinquity machinery on a compact carrier — a route since closed by Paper 45's degeneracy theorem). Both are valid compact approximations to the same physical Lorentzian Dirac.

A numerical-verification panel (Sprint L3c-α.2, 2026-05-23) at 24 cells $(n_{\max}, N_t) \in \{2, 3\} \times \{3, 5, 7, 11\}$ along three scalings ($T_0 = 2\pi$ canonical, $T_1 = N_t/\log N_t$ sub-linear, $T_2 = \sqrt{N_t}$ square-root) confirms Λ^L bit-exact T-and-N_t-independence at fixed $n_{\max}$: $\Lambda(n_{\max} = 2) = 2.074551$ and $\Lambda(n_{\max} = 3) = 1.610060$ to 6+ significant figures across all three scalings, matching Paper 45 Table 1 verbatim. Structural reason: the panel quantity only ever contained the spatial rate formula (the degeneracy analysis below), so it equals $\gamma^{\mathrm{SU}(2)}(n_{\max})$ exactly. (An earlier version attributed this to the joint rate $\max(\gamma^{\mathrm{SU}(2)}, \gamma^{U(1)})$, which at $N_t = 1$ is not $\gamma^{\mathrm{SU}(2)}$ in general.) The empirical confirmation confirms the T- and N_t-independence of the surviving *spatial* rate formula. Under the degeneracy analysis above this is an automatic formula identity (the panel quantity only ever contained the spatial rate formula), not an operator-level convergence, so it does not upgrade the descoped L3c-α panel to an established Lorentzian result.

9.3 Pythagorean orthogonality formal proof (Paper 42 §11.7)

The former Paper 43 (open question O4) named a formal proof of the diagonal/off-diagonal subspace-decomposition mechanism (a mechanism since withdrawn: the spatial wedge Dirac is chirality-diagonal, and the proof is a chirality-parity identity) behind the L2-F.1 (2026-05-17) empirical observation $\langle H_{\mathrm{local}}, D_L^W \rangle_{\mathrm{HS}} = 0$ as a follow-on. The formal proof landed on 2026-05-23 via a $\mathbb{Z}_2$ wedge-chirality parity argument and is now recorded in Paper 42 §11.7 (Theorem 11.14) as a formal theorem.

The argument has three structural inputs from Papers 42/44 [34, 35]: (I1) the wedge chirality grading Π_W (BBB Krein-chirality γ^5 restricted to the wedge) is a unitary self-adjoint involution; (I2) $[\Pi_W, H_{\mathrm{local}}] = 0$ (even parity of the posited BW-α generator $K_\alpha^W = \mathrm{diag}(2m_j)$ under wedge chirality); (I3) the wedge-restricted Lorentzian Dirac D_L^W has *mixed* parity — its temporal factor is Π_W-*odd* ($\{\gamma^0, \gamma^5\} = 0$) while its spatial factor D_{GV}^W is Π_W-*even* but chirality *sign-paired* (D_{GV} is chirality-diagonal — the same fact that makes the BBB universal axiom $\{\chi, D\} = 0$ *fail* on the truthful D_{GV} — so $D_{\mathrm{GV}}^W = \Pi_W |D_{\mathrm{GV}}^W|$ with $|D_{\mathrm{GV}}^W|$ even and χ-independent).

The orthogonality follows from two cyclicity-of-trace lemmas: (a) for any unitary involution Π, $[\Pi, A] = 0$ and $\{\Pi, B\} = 0$ imply $\mathrm{Tr}(A^\dagger B) = 0$ via $\mathrm{Tr}(A^\dagger B) = \mathrm{Tr}(A^\dagger \cdot (-B)) = -\mathrm{Tr}(A^\dagger B)$ (this kills the temporal factor); (b) for A even and χ-independent and $B = \Pi|B|$ even and sign-paired, $\mathrm{Tr}(A^\dagger B)$ cancels pairwise across the $\chi = \pm 1$ partners (this kills the spatial factor). The bit-exact L2-F.1 numerical panel and the 7-cell regression test are empirical corroboration of the now-algebraic identity at machine precision. (An earlier one-line "D_L^W is Π_W-odd" statement was corrected to this mixed-parity form, 2026-06-23, after the chirality-pairing test showed D_{GV}^W is chirality-diagonal hence Π_W-even.)

9.4 Master theorem of the math.OA arc

Paper 39 §6 (iii) and (v) named two open follow-ons: (iii) the k-fold extension of the Paper 39 two-factor tensor-product state-space GH bound to $\mathcal{T}_{S^3}^{\otimes k}$, and (v) the higher-rank tensor product $\mathcal{T}_{G_1} \otimes \mathcal{T}_{G_2}$ at arbitrary compact connected Lie group factors. Both were addressed on 2026-05-23 as *sketches* (only the $k = 2$ case is proved; see below).

(iii) k-fold extension — constant 1 (sketch). On the k-fold module with k mutually anticommuting Clifford generators, the compression argument of Paper 39 Lemma L3-T gives $C_3^{(k)} = 1$ for every k (Paper 39 §6(iii)); the operator-norm triangle bound with Cauchy–Schwarz gives only $\sqrt{k}$, and an earlier per-cutoff factor $\sqrt{(N_* - 1)/(N_* + 1)}$ is withdrawn with Paper 38's per-harmonic bound. *Correction (2026-06-18):* an earlier draft claimed a k-**independent** constant $\sqrt{(N_* - 1)/(N_* + 1)} \to 1^-$ via a "Pythagorean Leibniz" sum-of-squares identity $\|\sum_j X_j\|_{\mathrm{op}}^2 = \sum_j \|X_j\|_{\mathrm{op}}^2$. That identity is **false** (pairwise anticommutation of the $D^{(k)}$ summands does not make the commutator terms operator-norm orthogonal — verified maximally violated at $k = 2$ on the actual harmonics); the k-independent constant now comes from compression, not from orthogonality. Only the $k = 2$ case is proved (Paper 39); $k > 2$ is a sketch.

*(Upgrade, 2026-09-07 — the proved case is **stronger** than this paragraph implies.)* Paper 39 does not prove $k = 2$ through the joint Lipschitz constant at all: its theorem is the *lifted-state* assembly, which uses neither a Berezin map nor a partial inverse, and the paper states that $C_3^{(2)}$ (now 1, by compression) "belongs to the *abandoned* (B, P)-pair route and is not a rate constant of this theorem." Recorded because this is the *mirror* of the usual staleness: the owner strengthened its claim and this summary kept the weaker form — nothing is misspelt and nothing is retracted, so neither C16 nor a `cited_by` list can see it.

(v) Master theorem — compact-Lie-group factors (sketch). Combining the k-fold extension with Paper 40 [33] (general-G convergence conditional; the leading constant beyond SU(2) open) gives the expected multi-factor form (Paper 39 §6(v))

$$d_{\mathrm{GH}}^{(k)}\big(\mathcal{T}_{n_1,\ldots,n_k}^{(k)},\ \mathcal{T}_{G_1\times\cdots\times G_k}^{(k)}\big) \ \le\ \sum_{j=1}^{k} \gamma_{n_j}^{G_j}, \qquad C_3^{(k)} = 1, \tag{26}$$

for arbitrary $k \ge 1$ and compact connected simple Lie groups $G_1, \ldots, G_k$ with bi-invariant metrics (Paper 40's class; a central torus factor is excluded — Paper 40 asserts no value for $c(G_{\mathrm{ss}} \times T^k)$). The k-independent constant 1 comes from compression (each G_j factor by Paper 40's left-equivariance argument). The leading constant of each summand γ^{G_j} is open beyond SU(2) (Paper 40); and this multi-factor / general-G form is a **sketch** (proved only for $k = 1$, SU(2) and $k = 2$, SU(2) × SU(2), on the lifted-state route with leading constant 1; $k = 1$ for general G is conditional; $C_3^{(k)}$ is the approximation-pair route's constant, not a rate constant).

The master-theorem sketch subsumes Papers [31] ($k = 1$, $G = \mathrm{SU}(2)$), [33] ($k = 1$, general G, conditional), and the two-factor case treated above (§4.8, $k = 2$, $G_1 = G_2 = \mathrm{SU}(2)$), each of which proves a bound at most the displayed sum. Paper 45's product-carrier proposition at $S^3 \times S_T^1$ (`prop:product_action_seminorm`; signature-agnostic and not a Lorentzian claim — its earlier compression-device convergence theorem is withdrawn) is a $k = 2$, $G_1 = \mathrm{SU}(2)$, $G_2 = \mathrm{U}(1)$ product with per-factor cutoffs, which converges at the additive rate $\gamma^{\mathrm{SU}(2)} + \gamma^{\mathrm{U}(1)}$ (Paper 45, `eq:product_rate`); it lies outside Paper 40's semisimple class and is not an instance of the master theorem's universal constant.

10 Further closures: Krein–Mondino–Sämann bridge, OSLPLS, and closed-form continuum CFT-on-sphere values

The arc admits three further closures landed subsequent to the May 22, 2026 drafting of this synthesis. These add three further papers (48, 49, 50) and extend the math.OA standalone count from eleven to fourteen.

10.1 Krein-pointed proper QMS and the Mondino–Sämann bridge (Paper 48)

Status: the metric-level content of this subsection is descoped by Paper 45's degeneracy theorem; the bridge design survives as a conditional proposal.

Paper 48 [40] constructs Krein-pointed proper quantum metric spaces $\mathbf{KreinMetaMet}_{\mathrm{pp}}$ as the natural Krein-lift of Latrémolière's recent pointed proper quantum metric space framework [49], and builds a *bridge functor* $W : \mathbf{KreinMetaMet}_{\mathrm{pp}} \to \mathbf{LorPLG}_{\mathrm{cov}}$ to the Mondino–Sämann synthetic Lorentzian pre-length space category [17, 18]. The bridge is identified via the Connes–Rovelli thermal-time hypothesis [48]: the modular distance ^{K} on the Krein side measures thermal time (additive, forward triangle), while the proper-time distance ℓ^{L} on the Mondino–Sämann side measures geometric proper time (super-additive, reverse triangle); Connes–Rovelli identifies these as two readings of the same wedge KMS state, related by Wick rotation. The bridge is functorial, not isometric.

Conditional on the bridge construction — whose K^{+}-weak-form input is descoped by Paper 45's degeneracy theorem — Bridge Theorem 6.4 with all four properties (B1)+(B2)+(B3)+(B4) would yield seven newly accessible theorems on the GeoVac hemispheric wedge, including: (T3) bit-exact panel inheritance — the panel of bound values $\approx 2.0746, 1.6101, 1.3223$ at $(2,3)$, $(3,5)$, $(4,7)$ (values of the bound, not computed distances) of Paper 45 would transport verbatim; (T5) synthetic Pythagorean orthogonality with the $1/\pi^2$ prefactor (the posited-normalisation $1/\beta^2$ of Paper 42 §11.7, not a computed M_1 output; re-tiered 2026-09-28) as the inverse-squared bridge cover slab-period; and (T6) a proposed G2 metric-level *route* for the GeoVac wedge specifically, via the bridge $\to$ MS pLGH workaround. This last item is the most substantive design content of Paper 48: it offers a conditional synthetic-side workaround for the Paper 45 §1.4 G2 metric-level problem (contingent on the descoped bridge, hence a proposed route rather than a closure), sidestepping the absence of a published non-compact Latrémolière propinquity by working through the Mondino–Sämann category instead.

Paper 48 is the tenth math.OA standalone in the arc, sibling to Papers 38, 42, 43, 44, 45, 46, 47, 29, 32, 40.

10.2 OSLPLS and the strong-form bridge: twin paradox as quantum information (Paper 49)

Status: the Λ-inheritance and K^{+}-framed claims of this subsection are descoped by Paper 45's degeneracy theorem; the cocycle-deficit algebra and the OSLPLS category design survive.

Paper 49 [41] closes Paper 48 §8.1's open question Q1′ by constructing an *OSLPLS* (Operator-System Lorentzian Pre-Length Space) category and a strong-form bridge functor $W^{\mathrm{flip}} : \mathbf{KreinMetaMet}^{\mathrm{flip}}_{\mathrm{pp}} \to \mathbf{OSLPLS}_{\mathrm{cov}}$ on the Paper 46 Appendix B enlarged substrate (chirality-flipping generators $M^{\mathrm{flip}} = \mathrm{diag}(W, -W)$ where $\{J, M\} = 0$). The closure rests on a genuinely new operator-algebraic construction: the Connes–Rovelli thermal-time stack across *distinct* KMS states, extending Connes–Rovelli [48] from the single-KMS-state setting to multi-state stack consistency. Properties (B1′), (B2′) and the non-metric reading of (B3′) hold theorem-grade (Paper 49's Status note descopes the metric-ball reading of (B3′)); (B4′), the metric-level Λ-inheritance, is *descoped* (the Krein-side seminorm is degenerate, Paper 45/46).

The headline structural finding is the *twin-paradox-as-quantum-information* reading: strict super-additivity of the OSLPLS reverse triangle on off-orbit triples is the operator-algebraic dual of the special-relativistic twin paradox, with the deficit quantified by Datta's max-divergence chain inequality [50]. The detoured worldline pays *two* max-divergence cocycle costs $\Delta_{\max}S^{\sigma_1\to\sigma_2}+\Delta_{\max}S^{\sigma_2\to\sigma_3}$; the direct worldline pays *one* cost $\Delta_{\max}S^{\sigma_1\to\sigma_3}$; the strict inequality follows from Datta 2009's max-divergence chain inequality (the Umegaki relative entropy itself does not satisfy a chain inequality in general — see Paper 49 §5.4 and Q3′ for the honest scope). Numerical verification at $(n_{\max}, N_t) \in \{(2,3),(3,5),(4,7)\}$ confirms: Λ bit-exact at all three panel cells (rate-formula consistency, not a metric convergence), *illustrative* Umegaki relative-entropy deficits 66.998 / 68.720 / 81.256 nats positive and monotone-increasing in $n_{\max}$ (the load-bearing $D_{\max}$ chain is verified 96/96), Riemannian-limit recovery at $N_t = 1$ bit-exact.

The Bousso–Chandrasekaran–Rath–Shahbazi-Moghaddam 2020 "kink transform" / Connes cocycle flow gravity dual [47] appears here as a natural bulk reading of the OSLPLS construction (Paper 49 §11): the strict super-additivity of the OSLPLS reverse triangle on off-orbit triples translates, under the kink-transform dictionary, to a positive relative entropy on the gravitational dual — recovering the BCFM positivity statement from the operator-algebraic side.

Paper 49 is the eleventh math.OA standalone in the arc. Closed-form expression for the cocycle entropy production deficit as a function of OSLPLS state-space coordinates remains a named open question (Paper 49 §10); the genuine non-commutative Mondino–Sämann pre-length-space extension is the separate named open question Q2′, partially addressed by the OSLPLS containment (Paper 49 §10).

10.3 Closed-form CFT-on-sphere partition functions and their orthogonal $\log 2$ / $\zeta(3)/\pi^2$ decomposition (Paper 50)

Paper 50 [42] evaluates the standard zeta-regularised CFT-on-sphere free energies with the framework's closed Hurwitz-zeta forms (used throughout Papers 38–47 for spectral sums on the Camporesi–Higuchi tower), applied to the full, untruncated continuum spectra — the same sums Klebanov–Pufu–Safdi evaluate. The finite cutoff $n_{\max}$ never enters, and the state-space GH convergence of the truncations plays no role in the values: this is an independent closed-form re-derivation of known continuum values, via a combination of $\zeta'(-2)$, $\zeta'(-4)$, $\zeta(3)$, $\zeta(5)$, $\log 2$, and π identified symbolically or by high-precision PSLQ (digit counts as reported in Paper 50).

Closed-form match on S^3. For the conformally coupled scalar and Camporesi–Higuchi Dirac on round S^3,

$$F^{S^3}_{\text{scalar}} = \frac{\log 2}{8} - \frac{3\zeta(3)}{16\pi^2} \approx 0.0638 \qquad \text{(PSLQ integer relation } [8,2,-3]),$$
$$F^{S^3}_{\text{Dirac}} = \frac{\log 2}{4} + \frac{3\zeta(3)}{8\pi^2} \approx 0.21896 \qquad \text{(symbolic Hurwitz expansion)}.$$

These agree with the Klebanov–Pufu–Safdi values (the Dirac value symbolically, the scalar value to 61 digits with a PSLQ identification) [44], which are prior art. The framework has no bulk, so none of this is holographic (Paper 50's closing remark).

Orthogonal decomposition and log 3 cancellation theorem on S^5. On the (scalar, Dirac) sector, the continuum values decompose orthogonally:

$$F_D + 2F_s = \frac{\log 2}{2} \quad (\log 2 \text{ sector}), \qquad F_D - 2F_s = \frac{3\zeta(3)}{4\pi^2} \quad (\zeta(3)/\pi^2 \text{ sector}).$$

Following Paper 18's period-ring stratification, $\log 2$ is a level-2 cyclotomic period in M_3's ring (entering through the 2^s factor of the half-integer Hurwitz identity) and $\zeta(3)/\pi^2$ is an $M_1 \times M_3$

cross-product; no Seeley–DeWitt (M_2) content appears. An earlier labelling of $\log 2$ as M_2 is withdrawn (2026-09-29). On S^5, the pattern extends: $F^{S^5}_{\text{scalar}} = -\log(2)/128 - \zeta(3)/(128\pi^2) + 15\zeta(5)/(256\pi^4)$ (PSLQ relation $[32, -2, -2, 15]$), $F^{S^5}_{\text{Dirac}} = -(3\log 2)/128 - 5\zeta(3)/(128\pi^2) - 15\zeta(5)/(256\pi^4)$. The $\log 3$ contributions from the $(2/3)^s$ term of the half-integer Hurwitz function $\zeta_H(s, 5/2)$ in the Dirac spectral zeta cancel exactly through the Dirac multiplicity polynomial (Paper 50's $\log 3$ cancellation theorem; $\log 3$, a level-3 cyclotomic period, would lie outside M_3's level-≤ 4 ring), restricting the substrate-natural ring to $\log 2 \cdot \mathbb{Q} + \zeta(3)/\pi^2 \cdot \mathbb{Q} + \zeta(5)/\pi^4 \cdot \mathbb{Q}$ on S^5.

S^7–S^{11} ladder closure. Every odd rung $S^3, \ldots, S^{11}$ has a rational-coefficient closed form in $\mathcal{R}_d$ (Paper 50, Rem. general_d_conjecture: symbolic at S^3, S^5; measured by PSLQ at S^7–S^{11}, the printed S^7 and S^{11} scalar relations regression-tested).

Cataloguing the master Mellin engine in physics. The S^3 and S^5 closures are recorded in Paper 50 §9 as the first two entries of a running catalogue of master-Mellin-engine appearances on the physics side. Paper 50 also flags structural correspondences to M_2-sector results in the QFT literature, including Henningson–Skenderis holographic Weyl anomaly [45] (M_2 Seeley–DeWitt content), TT-bar / cutoff AdS [46], and the Chamseddine–Connes spectral action [4] (also M_2). These correspondences are flagged as structural, not as new derivations.

Structural reading: scale-freedom on substrate ↔ scale-invariance in physics. The closed-form CFT values on S^3 and S^5 are computed from the continuum spectra, so they are not evidence from the discrete substrate; what follows is a parallel, not an explanation. The GeoVac label lattice carries the diagonal SO(4) Casimir with pure-integer eigenvalues $n^2 - 1$, the multiplicities are integers, the Wigner-symbol couplings are exact rationals, and the packing diameter d_0 *cancels out* of every spectral quantity. Physical scales enter only through the Fock projection focal length p_0, which is a stereographic conformal map. This scale-freedom on the substrate is the geometric reflection of the same principle that conformal field theory [44] embodies on the physics side: a CFT has no preferred length scale at the level of its action and observables; the GeoVac graph has no preferred length scale at the level of its eigenvalues and matrix elements. This is *not* a claim that GeoVac *is* a CFT, nor that the substrate explains the Paper 50 values (they are continuum zeta determinants); it is a structural parallel only. (The substrate-side account of this principle is developed in the Group 3 foundations synthesis [43].)

Paper 50 is the twelfth math.OA standalone in the arc.

11 Open questions and outlook

On the Riemannian / spatial side Paper 38 proves convergence for the commutative (S^3) factor, unconditionally; the almost-commutative extension and Dirac-seminorm convergence are open. On the modular-Hamiltonian side, Paper 42 (including its § 11, the archived Paper 43's surviving content) records automatic finite-cutoff period closures of a posited generator on Riemannian and Krein backgrounds; deriving that generator from a Camporesi–Higuchi wedge state is open (2026-09-28). The Lorentzian *convergence* question is left open by Paper 45's annihilation theorem. Several questions remain open.

Strong-form Lorentzian propinquity (Q1) — OPEN (Paper 46 descoped, §9.1). Strong-form Lorentzian propinquity — a Latrémolière-style metric on Krein spectral triples in their own right, without K^+-restriction (Paper 45 §1.4 G1) — remains the principal open NCG-math problem of the Lorentzian arc. Paper 46 [37] proposed a closure via the operator-norm Lipschitz seminorm $L_{\text{op}}(a) = \|[D_L, a]\|_{\text{op}}$ on the chirality-doubled scalar-multiplier substrate of

Paper 44, but that seminorm has kernel far exceeding the scalars (the temporal multipliers are L_{op}-invisible), so the construction is not a quantum-metric distance and is descoped pending repair; the "free-upgrade" bit-exact panel match is a re-evaluation of the spatial rate formula, not an independent Lorentzian convergence. The enlarged-substrate algebra of Paper 46 Appendix B (chirality-flipping generators with $\{J, M\} = 0$, where the commutators are non-zero) survives and is the named starting point for the repair. A genuine strong-form theorem requires an indefinite-inner-product Lipschitz seminorm compatible with Krein-self-adjoint D_L; a substantial open program.

De-compactification $T \to \infty$ (G2) — CLOSED at norm-resolvent level by Paper 47 (§9.2); G2-metric open as a named frontier. The temporal factor $\mathrm{U}(1)_T$ is finite; the standard physics applications (Hawking, Unruh) require non-compact $\mathbb{R}_t$. The rate $T \log N_t/N_t \to 0$ in (21) is the joint mass-concentration on $\mathrm{U}(1)_T$ as $N_t \to \infty$ at fixed T; the $T \to \infty$ limit is closed at the *norm-resolvent / spectral* level by Paper 47 [38]'s two-rate hybrid composite (25). The remaining open question is the *metric / propinquity* level closure on a non-compact carrier (G2-metric or equivalently G1-genuine on the non-compact substrate), which requires a non-compact extension of Latrémolière's propinquity not in the published literature as of May 2026. This is named open frontier (Sprint L3e in our internal taxonomy), with the path-of-least-resistance being a pointed-propinquity bridge to the Mondino–Sämann synthetic Lorentzian GH program (§9.2).

Cross-manifold $\mathcal{T}_{S^3} \otimes \mathcal{T}_{\text{Hardy}(S^5)}$ (G3, Coulomb-side claim only). The natural tensor product of the Coulomb S^3 spectral triple with the Bargmann–Segal Hardy-sector spectral triple on S^5 [28] is blocked at the NCG-framework level by the Coulomb/HO asymmetry recorded in Paper 24 §V (seven layers catalogued, five standing; its first two layers, a spectrum-computing role for L_0 and calibration π, are withdrawn). The standing layers that bear on it include: (iii) non-abelian Wilson gauge with natural matter coupling (both sides have Wilson, only Coulomb-side has natural matter coupling because the Bargmann tower $(N, 0)$ irreps have N-dependent dimensions and no uniform-rep matter sector); and (iv) Pythagorean orthogonality of the posited wedge generator (Coulomb-side: the Krein wedge HS-orthogonality of Corollary 6.4 with closed-form prefactor $1/\beta^2$ of the posited normalisation; on the Hardy sector the natural analog operands are not orthogonal, **[ARGUED]**). Closing G3 requires new NCG-framework machinery to bridge Riemannian and Hardy-sector spectral triples in a single tensor product; we make no claim about whether this is achievable.

Higgs as inner fluctuation (G4). The GeoVac triple is naturally combined with a finite almost- commutative factor $\mathcal{A}_{\text{GV}} \otimes (\mathbb{C} \oplus \mathbb{H})$ to produce an electroweak-like sector. The Connes axiom audit at finite cutoff (Paper 32 §8.C) gives verdict POSITIVE-THIN: the construction *admits* a Higgs sector structurally (i.e. the inner fluctuation $D \mapsto D + \omega + \varepsilon' J\omega J^{-1}$ produces a non-trivial off-diagonal $\mathbb{C} \leftrightarrow \mathbb{H}$ block iff the finite Yukawa Y_F is non-zero), but *GeoVac does not autonomously select* Y_F from its internal data. In Marcolli–vS terms, GeoVac sits on the Yang–Mills-without-Higgs side of the Perez-Sanchez [20] clarification, with calibration data (the Yukawa) supplied externally. Paper 32 §8.C records this as the standing structural verdict.

State-side complement of the dictionary. Paper 44 highlights that the non-trivial K^+-positivity content of the Lorentzian operator system sits at the *state* level, not the operator level. The natural object is the Wasserstein–Kantorovich metric on K^+-states. Paper 45's K^+-weak-form analysis (the annihilation theorem) is the first probe in this direction and shows the operator-level compression is degenerate; substantially more mathematics on Wasserstein–Kantorovich on Krein-positive *states* is required to complete the state-side dictionary.

Concluding remark. The structural identification of GeoVac as an almost-commutative spectral triple in the Marcolli–van Suijlekom lineage rests, at the state-space Gromov–Hausdorff level, on Paper 38, which proves the commutative (S^3) factor only (convergence of the almost-commutative extension is **[OPEN]**); it is extended by Paper 39 (tensor product, state-space GH) and Paper 40 (compact connected semisimple Lie groups; for general G conditionally on the per-group spin-window decomposition). The Lorentzian K^+-weak-form extension is *descoped*: Paper 45 proves a degeneracy theorem (the natural Lorentzian-propinquity seminorm vanishes identically), and what survives on the temporal factor is Paper 45's signature-agnostic product-carrier rate (not a Lorentzian convergence). The Fejér-moment constant $4/\pi$ of the truncated triples' bound is read (a reading, not a derivation; Paper 38) as M_1 ($k = 0$ volume-slot) content; it is the kernel's constant, not the convergence rate, and the π-source case-exhaustion theorem (Theorem 8.1, **[CONDITIONAL]**) does not itself cover GH rates. (The $1/\pi^2$ of the Lorentzian Pythagorean closed form is the imported normalisation of a posited generator and is not shown to share that signature.)

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